

Pandora's Problem with Interdependent Values

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Abstract

We study sequential search when inspecting an alternative reveals its payoff and provides information about unopened alternatives. For a finite set under general dependence, optimal stopping is governed by a posterior reservation frontier: search continues when the incumbent lies below a scalar threshold conditional on beliefs and the remaining set. The frontier weakly exceeds the largest standalone reservation value, defining a learning premium, and search value is convex in beliefs. We give a sufficient condition for interval continuation rules and characterize their implications under additive interdependence. Log-concavity yields an interval when one box remains, while additional tail conditions ensure two finite cutoffs. Gaussian components yield two finite cutoffs after every inspection whenever continuation is possible. We also show when increasing common uncertainty reverses the preferred first inspection from a more dispersed to a more informative alternative. Two exponential specifications reveal further possibilities. With shortfalls from a common upper bound, an optimal inspection set can be chosen before observing any payoffs, despite Bayesian learning. With additions above a common lower bound, continuation, when possible, lies between two finite cutoffs even without strict posterior contraction.

Keywords: sequential search; Pandora's problem; interdependent values; Bayesian learning; reservation frontier; search order.

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1 Introduction

Search among related alternatives creates information spillovers. A consumer trying a product may learn about both that particular model and an unfamiliar technology shared by other products. A firm testing a production method may learn about both its implementation and the potential of the underlying technology. In either case, an inspection reveals a payoff that can be selected and changes expectations about alternatives that remain untried. When only total performance is observed, the decision maker must infer how much of a success or failure reflects the common component and how much is specific to the inspected alternative.

In the classical Pandora model of Weitzman (1979), independent values allow a simple solution to both stopping and search order. Each alternative has a reservation value determined by its own payoff distribution and inspection cost. Alternatives are inspected in descending reservation order, and search stops when the best available payoff reaches the highest remaining reservation value. With interdependent values, an inspection also changes the prospects of unopened alternatives. A poor realization can make further search unattractive by revealing bad news about their common component. A favorable realization can improve those prospects while simultaneously creating a better incumbent. These effects also change which alternative is most attractive to inspect: a useful source of information need not offer the best prospect of a high payoff.

This paper characterizes optimal stopping and inspection choices in this environment. There is a finite set of alternatives with values that are conditionally independent given an unobserved state. Inspection costs may differ, each inspection reveals the selected alternative's total value, and recall is perfect. An unrestricted latent state accommodates any finite joint distribution satisfying a mild integrability condition. At each history, the problem is summarized by the unopened set R , the posterior belief π , and the incumbent b .

Our general stopping result establishes a scalar *posterior reservation frontier* $\rho_R(\pi)$. Conditional on the posterior and the remaining set, search continues exactly when $b < \rho_R(\pi)$; the next inspection maximizes the expected payoff from opening a box and searching optimally thereafter. The key observation is that increasing the incumbent by one unit can raise the value of continuation by at most one unit. The net return to search is therefore nonincreasing in the incumbent, producing a threshold rule even under general dependence. More available alternatives or lower inspection costs weakly raise this frontier. The threshold belongs to the entire continuation problem and incorporates the information that inspections can provide about other alternatives.

The same frontier gives a sufficient condition for an interval rule in the realization of a newly inspected box. If the post-inspection frontier is nondecreasing in the observation and responds at a rate uniformly below one, continuation occurs on an interval whenever it occurs at all. Low realizations leave the incumbent fixed, while high realizations eventually raise it above the continuation frontier. This *posterior-contraction condition* applies to any remaining set and does not require additivity. It guarantees a finite upper cutoff, while the lower endpoint may be unbounded.

The general analysis also identifies two values of information. First, the posterior frontier weakly exceeds the highest reservation value obtained by evaluating each box in isolation. Their difference is a *learning premium*: the additional willingness to search arising from the option

to condition later inspections on an observed payoff. It is zero under independence or with a single remaining box, and we give a sufficient condition for strict positivity. Informative observations alone do not guarantee a positive premium, because the information must make some further inspection worthwhile. Second, search value is convex in posterior beliefs, so additional information supplied before search has nonnegative expected value. In particular, a Blackwell improvement in a preliminary experiment weakly raises expected search value when experiment costs are unchanged (Blackwell, 1953).

This comparison of preliminary experiments does not determine which box should be inspected, since each box also supplies a selectable payoff. A binary-state example makes the distinction explicit. The optimal first inspection switches between a more informative box and a more dispersed box as beliefs change, even though their conditional payoff distributions, costs, and the incumbent remain fixed. The example rules out a universally optimal ordering based on history-independent box indices.

We then study additive interdependence, where each value is the sum of a common component and an independent idiosyncratic component. When only one box remains after an inspection, log-concavity makes the gross gain from opening that box nondecreasing in the observation below the incumbent and nonincreasing above it. The continuation set is consequently an interval. Two finite endpoints require additional tail conditions. An exceptionally low payoff may be attributed mainly to idiosyncratic bad luck, leaving other alternatives attractive; an exceptionally high payoff may instead be attributed mainly to the common component, sustaining the value of further search. We identify sufficient conditions under which both tail gains vanish and give an example illustrating bounded intervals, half-lines, the empty set, and the whole real line as continuation sets.

Gaussian components extend the two-cutoff rule to any finite number of heterogeneous alternatives. Posterior variance depends on which boxes have been inspected, while the posterior mean responds linearly to the observed value. Translation invariance then makes the post-inspection frontier affine in that value, with slope strictly between zero and one whenever common uncertainty is positive. Thus every inspection that can induce further search has two finite realization cutoffs. Both are explicit functions of a scalar continuation frontier, which can be computed recursively. Holding beliefs and the incumbent fixed, more identical remaining boxes weakly raise the learning premium and widen the continuation interval. As common uncertainty vanishes, the frontier converges to the independent-value threshold and the usual Weitzman stopping rule is recovered.

The Gaussian model also permits an analytical comparison of search order. For two alternatives with the same mean and inspection cost, greater idiosyncratic dispersion increases the direct prize value, whereas lower dispersion makes an observation more informative about the common component. Under an explicit bound on inspection cost, we derive a threshold for the gap between the boxes' expected payoff and the incumbent that determines the preferred first inspection when common uncertainty is large. If that gap exceeds the threshold, the more dispersed box is uniquely preferred when the common component is known, but the more precise box is uniquely preferred for all sufficiently large common variance. The optimal order therefore reverses at least once. A precise observation is valuable in this case because an unfavorable surprise can identify states in which another inspection is not worth its cost. If the gap is below the threshold, the

more dispersed box is preferred for sufficiently large common variance.

Exponential components provide a different tractable specification with a natural interpretation in technology search. Suppose each method’s payoff is an unknown common potential less an independent exponential implementation loss. When the common potential has an exponential prior above the existing method’s payoff, an optimal inspection set can be chosen before any values are observed. A new record raises the incumbent and the posterior lower bound on common potential by the same amount. The distribution of the remaining distance to that potential therefore depends on which methods have been tested, but not on their realized payoffs. We obtain an explicit value for every inspection set and show that a minimal maximizing set can be inspected in any order. Simple priority rules follow when costs or exponential rates are common. Correlated payoffs and Bayesian learning thus coexist with an optimal plan whose inspection and stopping decisions do not depend on realized values.

A second exponential specification places each payoff above a common lower bound through an independent exponential addition. Here low observations reduce the posterior frontier, while sufficiently high observations leave beliefs unchanged and improve the incumbent. Whenever continuation is possible, it occurs between two finite cutoffs, expressed in terms of the remaining reservation frontier. The posterior frontier moves one-for-one with the observation over part of its support, so this example establishes the interval rule beyond uniform strict posterior contraction.

The paper builds on the optimal search literature initiated by Weitzman (1979). Related extensions allow selection without prior inspection (Doval, 2018) or study the acquisition of information about product attributes (Branco et al., 2012; Klabjan et al., 2014; Ke et al., 2016).

The closest economic work studies learning across related alternatives. Ke and Lin (2020) characterize search over products with uncertain common and idiosyncratic attributes and examine its implications for demand complementarity and price competition. Gambato (2024) develops a history-dependent scoring rule for infinitely many products arranged on a grid, where inspection reveals their shared attributes separately. Huh et al. (2025) combine learning about an alternative through repeated sampling with learning about the population from which alternatives are drawn, obtaining thresholds for accepting the current alternative or switching to a new one. We study a finite collection in which each inspection reveals a total payoff. Our lower stopping cutoff governs when bad news ends search over the remaining collection, and our Gaussian and exponential results characterize how that decision interacts with the available prize and the choice of the next inspection.

The algorithmic literature studies Pandora’s problem under unrestricted correlation. Chawla et al. (2020) develop learning and approximation results with sample access to the joint distribution. Chawla et al. (2023) relate the approximation of fully adaptive policies to decision-tree and covering problems, while Gergatsouli and Tzamos (2023) obtain approximation guarantees using Weitzman-type rules. Our contribution is an exact structural characterization of stopping, together with distributional specifications that yield explicit cutoff formulas and analytical comparisons of inspection choices.

Section 2 presents the model. Section 3 establishes the general stopping rule, interval conditions, and information results. Section 4 studies additive interdependence, covering the last unopened box under log-concavity in Section 4.1, Gaussian components in Section 4.2, and exponential components in Section 4.3. Section 5 concludes. Proofs are in the appendix.

2 The Model

There is a finite set of boxes $\mathcal{I} = \{1, \dots, N\}$. There is an underlying state $\theta \in \Theta$, drawn from a prior π_0 . The state is not directly observed by the decision maker. Conditional on θ , box values are independent, with box i distributed according to $P_i(\cdot \mid \theta)$. Its conditional cumulative distribution function is $F_i(v \mid \theta) = P_i((-\infty, v] \mid \theta)$, and its value is denoted by V_i . The decision maker knows the prior and all conditional distributions.

Opening box i costs c_i and reveals V_i . Search is sequential and recall is perfect. At any history, the decision maker may stop and select the best inspected value or a deterministic outside option $u_0 \in \mathbb{R}$. If S is the set of inspected boxes, terminal utility, net of all inspection costs paid along the history, is

$$\max \left\{ u_0, \max_{i \in S} V_i \right\} - \sum_{i \in S} c_i. \quad (1)$$

The decision maker may stop before opening any box. Relative to the classical Weitzman (1979) model, the key departure is that box values are linked through the common underlying state θ . Because θ is not directly observed, each inspection is informative not only about the inspected box but also about the remaining boxes: as search proceeds, the decision maker updates her posterior belief about θ , which in turn changes the predictive distributions of all unopened boxes.

For any real number x , write $x^+ = \max\{x, 0\}$; in particular, $V_i^+ = \max\{V_i, 0\}$. For a subset $R \subseteq \mathcal{I}$, a belief π over θ is *R-admissible* if

$$\mathbb{E}_\pi \left[\max_{i \in R} V_i^+ \right] < \infty, \quad (2)$$

where the maximum over the empty set is zero. In short, call a belief π *admissible* when it is $\mathcal{I}$ -admissible. The following assumption is made.

Assumption 1. *For every $i \in \mathcal{I}$, $c_i > 0$, and the prior π_0 is admissible.*

Assumption 1 is a mild regularity condition ensuring that the search problem is well defined. Admissibility bounds the expected positive part of the best attainable payoff and therefore guarantees that the value of every feasible search policy, and hence the continuation value functions introduced below is finite.

A *history* is denoted as $h = (S, v_S)$, where $S \subseteq \mathcal{I}$ is the inspected set and $v_S = (v_i)_{i \in S}$ is the vector of observations. Let

$$R(h) = \mathcal{I} \setminus S, \quad b(h) = \max \left\{ u_0, \max_{i \in S} v_i \right\} \quad (3)$$

denote the *unopened set* and the *incumbent*. Let π_h be the *posterior* belief of θ given h . Given a current belief π , for a fixed unopened box $i \in R$, let Q_i^π denote the *predictive belief* of V_i . Its cumulative distribution function is

$$Q_i^\pi(v) = \int_{\Theta} F_i(v \mid \theta) \, d\pi(\theta). \quad (4)$$

Based on this predictive belief, suppose the decision maker then opens the box i . A regular

conditional posterior after observing $V_i = v$ is denoted by $T_i(\pi, v)$, defined Q_i^π -almost surely. When conditional densities exist, Bayes' rule can be written as

$$dT_i(\pi, v)(\theta) = \frac{f_i(v \mid \theta)}{\int_{\Theta} f_i(v \mid \vartheta) d\pi(\vartheta)} d\pi(\theta). \quad (5)$$

Thus, Q_i^π captures the decision maker's prediction of the realization of box i before inspection, whereas $T_i(\pi, v)$ captures the posterior belief about θ after observing that realization. Together, they fully describe the Bayesian information-updating process in the sequential search problem. Moreover, conditional independence implies that the distribution of all unopened values given a history depends on the observations only through π_h . Past observations affect terminal choice only through $b(h)$. Thus, after any search history, all payoff-relevant information for the continuation problem is summarized by (R, π, b) , i.e. (R, π, b) constitutes a sufficient state.

For simplicity, define $x \vee y = \max\{x, y\}$ for any $x, y \in \mathbb{R}$. The *continuation value* $W_R(\pi, b)$, then satisfies $W_\emptyset(\pi, b) = b$ and for $R \neq \emptyset$,

$$W_R(\pi, b) = \max \left\{ b, \max_{i \in R} H_{i,R}(\pi, b) \right\}, \quad (6)$$

where

$$H_{i,R}(\pi, b) = -c_i + \int_{\mathbb{R}} W_{R \setminus \{i\}}(T_i(\pi, v), b \vee v) dQ_i^\pi(v). \quad (7)$$

Here, $H_{i,R}(\pi, b)$ is the expected payoff from opening box i immediately and then following an optimal search policy thereafter. For future convenience, define the best continuation value from unopened boxes as

$$H_R(\pi, b) = \max_{i \in R} H_{i,R}(\pi, b)$$

and the corresponding continuation surplus $G_R(\pi, b) = H_R(\pi, b) - b$. Throughout, whenever the decision maker is indifferent between stopping and continuing, we assume that she stops.

3 General Results

3.1 Optimal Rule

We now characterize the optimal stopping rule in the general environment. Fixing the set of unopened boxes and the current belief, the only remaining state variable relevant for stopping is the incumbent b . The key question is therefore how the value of search changes as the incumbent improves. Since stopping yields exactly b , a threshold stopping rule will emerge if the value of continuing cannot increase faster than the stopping payoff. The following proposition establishes precisely this property, together with several useful regularity properties of the value functions.

Proposition 1. *Fix an unopened set R , a belief π and an unopened box $i \in R$. The functions $b \mapsto W_R(\pi, b)$ and $b \mapsto H_{i,R}(\pi, b)$ are finite, continuous, nondecreasing, convex, and one-Lipschitz. In particular, for $b' \geq b$,*

$$0 \leq W_R(\pi, b') - W_R(\pi, b) \leq b' - b, \quad (8)$$

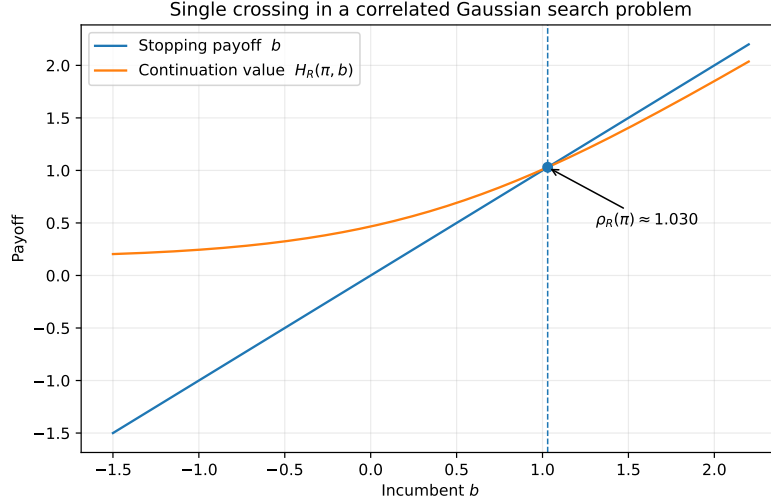


Figure 1: Single crossing between the continuation value and the stopping payoff. The calibration is $q = 1$, $\mu = 0$, $\sigma = 1$, and $c = 0.2$, with two unopened boxes.

and the same inequalities hold for $H_{i,R}$.

The most important property for the stopping problem is the one-Lipschitz bound. A higher incumbent can never reduce the decision maker's payoff, since the previous search policy remains feasible. At the same time, increasing the incumbent by one unit can increase the terminal payoff by at most one unit under every realization and every search policy. Hence the continuation value is nondecreasing in b , but it can never rise faster than the payoff from stopping.

The proof follows this intuition by backward induction on the number of unopened boxes. After opening box i and observing $V_i = v$, the incumbent becomes $b \vee v$. If b is increased to b' , the new incumbent changes by an amount between zero and $b' - b$. The induction hypothesis therefore implies that the continuation value after every realization changes by no more than $b' - b$. Taking expectations and then maximizing over feasible actions preserves this bound. Convexity follows similarly from the convexity of $b \mapsto b \vee v$ and the fact that integration and pointwise maximization preserve convexity, while finiteness follows from Assumption 1.

Economically, Proposition 1 says that a better incumbent weakens the relative attractiveness of further search. The value of continuing may rise with b , because the decision maker retains the improved outside option, but it rises no faster than the value of stopping. This single-crossing structure is the key step toward the general reservation-frontier characterization below.

Figure 1 illustrates the single-crossing property in a simple Gaussian environment. Consider two identical unopened boxes $V_i = X + \varepsilon_i$ with $X \sim N(0, 1)$, $\varepsilon_i \sim N(0, 1)$, and inspection cost $c = 0.2$. The continuation value $H_R(\pi, b)$ is increasing in the incumbent, but rises more slowly than the stopping payoff b . The two curves cross at $\rho_R(\pi) \simeq 1.030$: search is optimal below this value and stopping is optimal above it. The figure illustrates the key role of the one-Lipschitz property in generating a threshold stopping rule.

Proposition 1 immediately implies a single-crossing property between the value of continuing and the payoff from stopping. Since $H_R(\pi, \cdot)$ is one-Lipschitz, the continuation surplus $H_R(\pi, b) - b$

is weakly decreasing in the incumbent b . Thus, once stopping becomes optimal as the incumbent rises, it remains optimal thereafter. The next theorem shows that this single-crossing property yields a scalar threshold characterization of optimal stopping even under arbitrary dependence among box values.

Theorem 1 (Optimal Rule). *Under Assumption 1, for every nonempty R and every R -admissible posterior π , there exists a finite scalar*

$$\rho_R(\pi) = \inf \{b \in \mathbb{R} : H_R(\pi, b) \leq b\}. \quad (9)$$

The optimal rule is characterized as follows:

- (i) *if $b \geq \rho_R(\pi)$, stopping;*
- (ii) *if $b < \rho_R(\pi)$, continuing search the box (or one of the boxes) with the highest $H_{i,R}(\pi, b)$.*

We call $\rho_R(\pi)$ the *posterior reservation frontier*. Theorem 1 shows that although interdependence destroys the box-by-box separability of the classical Weitzman problem, it does not destroy threshold stopping. Conditional on the current posterior π and the remaining opportunity set R , the stopping decision is summarized by a single scalar: a sufficiently low incumbent induces further search, whereas a sufficiently high incumbent induces stopping.

The distinction from the classical Weitzman reservation value is important. The frontier $\rho_R(\pi)$ is not attached to an individual box. It depends jointly on the posterior belief and the entire set of remaining opportunities, because an inspection may change beliefs about every unopened box. Thus the frontier determines *when* to stop, whereas the choice of *which* box to inspect next remains state dependent and is governed by the expected continuation payoffs.

The frontier also admits simple comparative statics.

Proposition 2 (Comparative statics of the posterior frontier). *Fix a posterior belief π .*

- (i) *If $R \subseteq R'$ and π is R' -admissible, then $\rho_R(\pi) \leq \rho_{R'}(\pi)$.*
- (ii) *Lowering the inspection cost of any box weakly raises the posterior frontier.*

Both comparative statics follow from simple feasibility arguments. Expanding the opportunity set cannot reduce the value of continuation because the decision maker can always ignore the additional boxes. Similarly, lowering an inspection cost weakly increases the payoff from every policy that uses that box while leaving all other policies unchanged. In either case, continuation remains attractive up to a weakly higher incumbent, and hence the posterior reservation frontier weakly increases.

3.2 Sufficient Condition for Interval Rules

At an arbitrary search stage, the realization of an inspected box changes not only the incumbent but also the entire continuation problem through Bayesian learning. Let R be the unopened set before inspecting box $i \in R$, and suppose that $R' = R \setminus \{i\}$ is nonempty. For a realization $V_i = v$, define the post-signal reservation frontier

$$r_i(v) \equiv \rho_{R'}(T_i(\pi, v)).$$

By Theorem 1, after observing v the decision maker continues searching if and only if $b \vee v < r_i(v)$. Hence, the shape of the realization-contingent search rule is determined by how the posterior frontier $r_i(v)$ responds to the observed value.

A direct sufficient condition is that favorable realizations raise the continuation frontier, but less than one-for-one. Specifically, suppose that r_i is continuous and, for some $\kappa_i \in [0, 1]$,

$$0 \leq r_i(v') - r_i(v) \leq \kappa_i(v' - v) \quad \text{for all } v' > v. \quad (\text{PC})$$

We refer to (PC) as the *posterior-contraction condition*.

Theorem 2. *Suppose (PC) holds. If $r_i(b) \leq b$, no realization of box i induces further search. If $r_i(b) > b$, there exist interval rules that continue searching if and only if $\underline{v}_i < v < \bar{v}_i$, where $\underline{v}_i$ may equal $-\infty$ whereas $\bar{v}_i$ is finite. The upper cutoff is the unique solution to $r_i(\bar{v}_i) = \bar{v}_i$, while the lower cutoff satisfies $\underline{v}_i = \sup\{v \leq b : r_i(v) \leq b\}$.*

The condition has a simple geometric interpretation. For $v \leq b$, the incumbent remains fixed at b , whereas the posterior frontier is nondecreasing in the signal. Thus the decision can switch only from stopping to continuation as v rises. For $v \geq b$, the incumbent itself rises one-for-one with v , while the frontier rises at most at rate $\kappa_i < 1$. The incumbent therefore eventually overtakes the continuation frontier. Intermediate realizations are precisely those for which neither sufficiently bad news nor a sufficiently strong incumbent makes further search unattractive.

3.3 The Value of Information

The optimal rule in the previous subsection makes clear that information is central to the search problem: the stopping frontier $\rho_R(\pi)$ depends on the current posterior, and each inspection changes that posterior before subsequent decisions are made. This section characterizes the value of such information from two complementary perspectives. First, we compare the posterior frontier with the reservation value obtained when each box is evaluated in isolation. The gap captures the option value of using an inspection to guide later search decisions. Second, we ask how the overall value of search changes when the decision maker receives additional information about the underlying state before searching. Together, the two results separate the value of information generated *within* the search process from the value of information available *before* search begins.

The learning premium. For each box i , define its posterior one-box reservation value by

$$z_i(\pi) = \inf \{b \in \mathbb{R} : \mathbb{E}_{Q_i^\pi} [(V_i - b)^+] \leq c_i\}. \quad (10)$$

The object $z_i(\pi)$ evaluates box i in isolation: the decision maker opens i and stops immediately afterward, without using its realization to condition any further search. It is finite and uniquely characterized by

$$\mathbb{E}_{Q_i^\pi} [(V_i - z_i(\pi))^+] = c_i. \quad (11)$$

Thus, opening box i and then stopping is strictly profitable if and only if $b < z_i(\pi)$.

The one-box reservation value provides a natural benchmark for the posterior reservation frontier. The former captures only the direct prize value of inspecting a box, whereas the latter also incorporates the option to use the realized value to update beliefs and adapt subsequent search decisions. Based on our notations, Weitzman (1979) tells that

$$\rho_R(\pi) = \max_{i \in R} z_i(\pi). \quad (12)$$

when the boxes' values are independent of each other under belief π . The next Proposition states that the correlation information provides nonnegative value and tells when it becomes positive.

Proposition 3 (Learning premium). *For every nonempty R ,*

$$\rho_R(\pi) \geq \max_{i \in R} z_i(\pi). \quad (13)$$

*Equality holds when boxes' values are independent under π or R is a singleton. Accordingly, define the **learning premium** by*

$$\Lambda_R(\pi) = \rho_R(\pi) - \max_{i \in R} z_i(\pi) \geq 0. \quad (14)$$

Let $i^ \in \arg \max_{i \in R} z_i(\pi)$ and write $z^* = z_{i^*}(\pi)$. If*

$$\mathbb{P}(W_{R \setminus \{i^*\}}(T_{i^*}(\pi, V_{i^*}), z^* \vee V_{i^*}) > z^* \vee V_{i^*}) > 0, \quad (15)$$

then $\Lambda_R(\pi) > 0$.

The first inequality follows from a simple option-value argument. Opening a box and stopping immediately is always feasible in the full dynamic problem, so allowing the decision maker to condition future inspections on the observed realization can only increase the value of continuation. When only one box remains, this additional option is absent and the frontier coincides with the one-box reservation value. The independent-values case connects the result directly to Weitzman (1979). Under independence, observing one box does not change the distributions of the remaining boxes, and the posterior frontier reduces to the largest box-specific reservation value. Hence $\Lambda_R(\pi) = 0$.

Interdependence may make the inequality strict because an inspection can reveal information about the attractiveness of other boxes. The strictness condition in (15) formalizes exactly this possibility. At $b = z^*$, opening i^* and stopping yields exactly the same expected payoff as stopping immediately. If its realization generates, with positive probability, a posterior under which some further inspection becomes worthwhile, contingent continuation makes the initial inspection strictly profitable. The optimal stopping frontier must then satisfy $\rho_R(\pi) > z^*$.

The learning premium therefore measures one particular value of information in the search process: the additional willingness to search created by the ability to use one inspection to inform later inspection decisions. It is an endogenous information value, because the same action that produces the signal also produces a payoff that may ultimately be selected.

It is worth mentioned that the conditions above are sufficient but not necessary. Correlation and informational content alone do not imply a positive learning premium. For example, suppose two boxes are perfectly correlated and identically valued, $V_1 = V_2 = X$, with strictly positive

inspection costs. Opening one box fully reveals the value of the other, but this information has no continuation value because the second box offers exactly the same payoff and requires an additional positive cost; hence the learning premium is zero despite perfect correlation and full information revelation. Conversely, when boxes are correlated, the inspection that generates the learning premium need not be a box i^* with the highest one-box reservation value. A box with a lower standalone reservation value may be sufficiently more informative about the underlying state that inspecting it creates valuable contingent continuation opportunities. Thus, condition (15) is also not necessary for positive learning premium.

The value of prior information. A complementary question is whether information has value even when it is supplied separately from the search alternatives. Consider a preliminary experiment about θ that generates a random posterior Π before any further search decision is made. Bayes plausibility requires that the posterior average back to the current belief, $\mathbb{E}[\Pi] = \pi$. The next result shows that additional information of this kind can only increase expected search value.

Proposition 4 (Convexity in beliefs and the value of information). *For every unopened set R and incumbent b , the map $\pi \mapsto W_R(\pi, b)$ is convex. Hence, for any Bayes-plausible random posterior Π satisfying $\mathbb{E}[\Pi] = \pi$,*

$$\mathbb{E}[W_R(\Pi, b)] \geq W_R(\pi, b). \quad (16)$$

Consequently, among preliminary experiments with the same cost, a Blackwell-more-informative experiment weakly raises expected search value.

The intuition is the standard value-of-information argument. For any fixed search policy, expected payoff is linear in the belief about θ . The optimal value $W_R(\pi, b)$ is the supremum over all such policy-specific linear payoffs and is therefore convex in π . A more informative experiment creates a mean-preserving spread of posterior beliefs, allowing the decision maker to adapt her subsequent search policy more closely to the information received. Jensen's inequality then yields (16).

The two propositions capture distinct but related informational forces. The learning premium compares the full sequential-search problem with a benchmark that ignores the informational consequences of an inspection and therefore measures the value of *endogenous cross-alternative learning*. The convexity result instead isolates the value of a *pure information experiment* received before search. In both cases, information is valuable because it allows future search decisions to be conditioned on a more informative belief about the underlying state.

The Blackwell comparison above concerns information supplied separately from the search alternatives. An inspection also generates a selectable payoff, so greater informativeness may conflict with greater prize value. The following example illustrates how this tradeoff affects the optimal search order.

Example 1 (Information and prize value). *Let $\theta \in \{0, 1\}$ with current belief $\Pr(\theta = 1) = p \in (1/2, 1)$. There are two unopened boxes, $R = \{A, B\}$, with common inspection cost $c = 1/4$ and incumbent $b = 1/4$. Box A pays $V_A = \theta$, while box B pays 2θ or zero, each with probability $1/2$,*

conditional on θ . Thus A fully reveals the state, whereas observing $V_B = 0$ leaves it uncertain. Both boxes have mean p , but B offers a more dispersed prize and has a higher one-box reservation value.

If A is inspected first, search continues only after $V_A = 1$. If B is inspected first, search continues only after $V_B = 0$. Both orders therefore select the same terminal prize. Their difference lies in the probability of avoiding the second inspection: A permits stopping upon identifying the bad state, with probability $1 - p$, whereas B permits stopping upon finding the high prize, with probability $p/2$. Consequently,

$$H_{A,R}(\pi, b) - H_{B,R}(\pi, b) = c \left(1 - p - \frac{p}{2} \right) = \frac{2 - 3p}{8}.$$

Search is worthwhile throughout the specified range. The more informative box A is uniquely optimal to inspect first when $1/2 < p < 2/3$, while the less informative box B is uniquely optimal when $p > 2/3$; the two orders tie at $p = 2/3$. As favorable states become more likely, the opportunity to stop with a high prize outweighs the opportunity to stop after learning that the state is unfavorable.

Failure of history-independent box indices. Example 1 also rules out a universally optimal ordering based on fixed box-specific indices. The conditional payoff distributions and inspection costs remain unchanged, yet A is uniquely optimal to inspect first at $p = 3/5$, whereas B is uniquely optimal at $p = 4/5$. Both posteriors can arise from earlier observations that leave the incumbent and the two unopened boxes unchanged. Any fixed indices, together with history-independent tie-breaking, prescribe the same order at both histories and must therefore be suboptimal at one of them. This argument rules out history-independent rankings; it does not by itself rule out indices that depend on the current posterior.

4 Additive Interdependence

The posterior reservation frontier in Section 3 characterizes the stopping decision for an arbitrary dependence structure. It does not, however, describe how the continuation decision varies with the realization of a newly inspected box. We now impose additive interdependence,

$$V_i = X + \varepsilon_i, \tag{17}$$

and study this realization-contingent decision. Here, X is a common component and the idiosyncratic components $(\varepsilon_i)_{i \in \mathcal{I}}$ are mutually independent and independent of X . In this additive model, the underlying state θ introduced in the general framework is the common component X . The additive structure separates two forces generated by an observation: below the incumbent, the realization affects continuation primarily through beliefs about the common component; above the incumbent, it also raises the payoff that future boxes must beat. This interaction generates a natural two-sided stopping rule under certain conditions.

4.1 The Last Unopened Box

We first study the final continuation decision, where the general posterior frontier admits a particularly sharp characterization. Denote f_X and f_i , $i \in I$ as the density of the common component X and the idiosyncratic components. The following assumption is needed.

Assumption 2. *Densities f_X and f_i , $i \in I$, are positive, continuous, and log-concave on $\mathbb{R}$.*

This assumption is preserved under Bayesian updating (Milgrom, 1981; Bagnoli and Bergstrom, 2005). After any finite history $h = (S, v_S)$, the posterior density of X is proportional to $f_X(x) \prod_{k \in S} f_k(v_k - x)$, which is again positive, continuous, and log-concave.

Suppose box i is inspected and only box j remains unopened. Let b be the incumbent before inspecting i and let $V_i = v$. Since the continuation problem now contains a single box, Proposition 3 gives $\rho_{\{j\}}(T_i(\pi, v)) = z_j(T_i(\pi, v))$. Hence the decision maker continues if and only if $b \vee v < z_j(T_i(\pi, v))$. Equivalently, define the gross *continuation gain*

$$g_{ij}(v; b) = \mathbb{E}[(V_j - b \vee v)^+ | V_i = v] = \begin{cases} \mathbb{E}[(X + \varepsilon_j - b)^+ | X + \varepsilon_i = v], & v \leq b, \\ \mathbb{E}[(\varepsilon_j - \varepsilon_i)^+ | X + \varepsilon_i = v], & v \geq b. \end{cases} \quad (18)$$

Box j is inspected precisely when $g_{ij}(v; b) > c_j$. Write $\mathcal{K}_{ij}(b, c_j) = \{v : g_{ij}(v; b) > c_j\}$ for the corresponding continuation set.

Proposition 5. *Under Assumption 2, $g_{ij}(\cdot; b)$ is continuous, nondecreasing on $(-\infty, b]$, and nonincreasing on $[b, \infty)$. Consequently, $\mathcal{K}_{ij}(b, c_j)$ is an interval. It is empty if $c_j \geq g_{ij}(b; b)$ and contains b if $c_j < g_{ij}(b; b)$.*

The two sides of the proposition reflect distinct effects of the signal. When $v \leq b$, the new box does not replace the incumbent, so a higher realization matters only through beliefs: log-concavity implies that the posterior of X shifts upward in monotone likelihood ratio order, making the remaining box more attractive. When $v \geq b$, the incumbent itself moves with v . Additivity is important here: conditional on $X + \varepsilon_i = v$, the comparison with the remaining box reduces to $V_j - v = \varepsilon_j - \varepsilon_i$. A higher realization shifts the conditional distribution of ε_i upward and therefore lowers the value of further search. The monotonicity is strict whenever the posterior distribution shifts strictly in monotone likelihood ratio order.

The interval result does not by itself imply two finite cutoffs. Define the two tail gains

$$g_{ij}^-(b) = \lim_{v \rightarrow -\infty} g_{ij}(v; b), \quad g_{ij}^+(b) = \lim_{v \rightarrow +\infty} g_{ij}(v; b). \quad (19)$$

When $c_j < g_{ij}(b; b)$, the lower endpoint of the continuation interval is finite if $c_j > g_{ij}^-(b)$ and is $-\infty$ if $c_j < g_{ij}^-(b)$; symmetrically, the upper endpoint is finite if $c_j > g_{ij}^+(b)$ and is $+\infty$ if $c_j < g_{ij}^+(b)$. At equality, boundedness depends on whether the corresponding tail limit is attained; under strict monotonicity, the endpoint is infinite.

Example 2. *Let $X \sim AL(3, 1)$, $\varepsilon_i \sim AL(1, 3)$, $\varepsilon_j \sim AL(2, 2)$, independently, where the asymmet-*

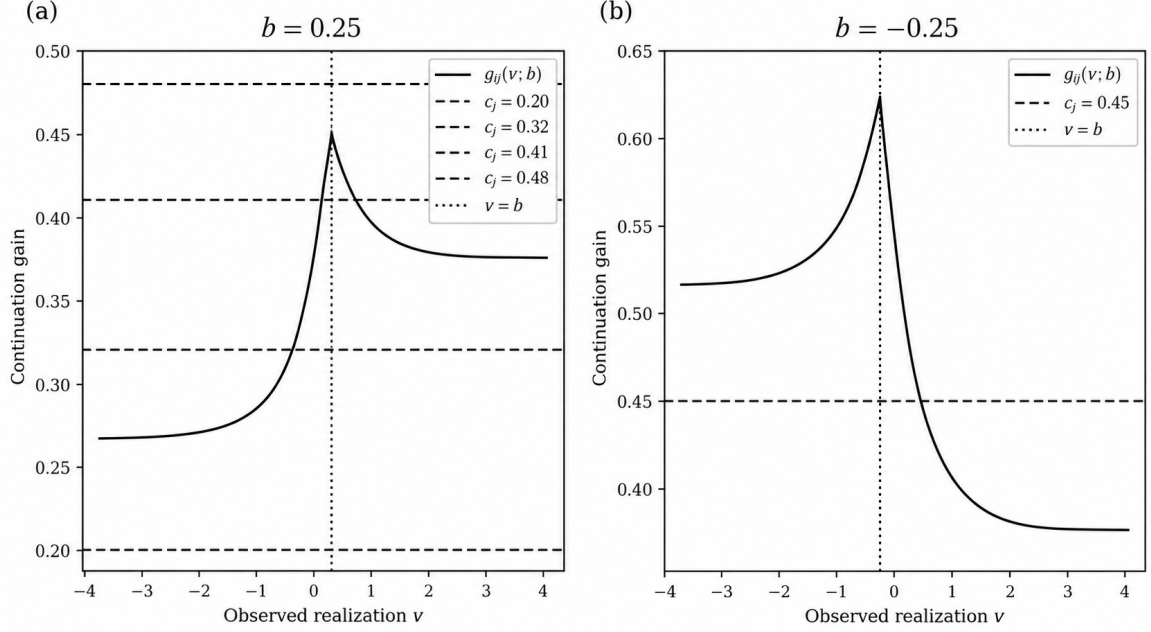


Figure 2: Continuation Gain and Interval Rules

ric Laplace density is

$$h_{\lambda_-, \lambda_+}(x) = \frac{\lambda_- \lambda_+}{\lambda_- + \lambda_+} \begin{cases} e^{\lambda_- x}, & x < 0, \\ e^{-\lambda_+ x}, & x \geq 0. \end{cases}$$

These densities are positive, continuous, and log-concave, so Proposition 5 applies. Figure 2 plots the continuation gain $g_{ij}(v; b)$. In panel (a), with $b = 0.25$, the tail limits satisfy $g_{ij}^-(b) = 0.265$, $g_{ij}^+(b) = 0.375$, and $g_{ij}(b; b) \approx 0.456$. Hence varying the search cost c_j generates four cases: $\mathcal{K}_{ij}(b, c_j) = \mathbb{R}$ if $c_j < g_{ij}^-(b)$, $\mathcal{K}_{ij}(b, c_j) = (\underline{v}, \infty)$ if $g_{ij}^-(b) < c_j < g_{ij}^+(b)$, $\mathcal{K}_{ij}(b, c_j) = (\underline{v}, \bar{v})$ if $g_{ij}^+(b) < c_j < g_{ij}(b; b)$, and $\mathcal{K}_{ij}(b, c_j) = \emptyset$ if $c_j \geq g_{ij}(b; b)$. In panel (b), only the incumbent changes, to $b = -0.25$. Then $g_{ij}^-(b) = 0.515$ and $g_{ij}^+(b) = 0.375$, so for any $c_j \in (g_{ij}^+(b), g_{ij}^-(b))$ the continuation set is $\mathcal{K}_{ij}(b, c_j) = (-\infty, \bar{v})$. Together, the two panels illustrate all possible shapes of the continuation set.

Thus monotone Bayesian updating is sufficient for the *interval* shape, but not for two-sided stopping with finite thresholds. A convenient sufficient condition for both tail gains to vanish is

$$X \mid (X + \varepsilon_i = v) \rightarrow -\infty \text{ as } v \rightarrow -\infty, \quad \varepsilon_i \mid (X + \varepsilon_i = v) \rightarrow +\infty \text{ as } v \rightarrow +\infty, \quad (20)$$

where convergence is in probability, together with uniform integrability of the corresponding positive parts in (18). The first condition attributes sufficiently bad realizations to a sufficiently low common component; the second attributes sufficiently good realizations partly to a sufficiently high idiosyncratic component.

Corollary 1. *Under Assumption 2 and (20), together with the stated uniform-integrability condition, $g_{ij}^-(b) = g_{ij}^+(b) = 0$. Hence, whenever $0 < c_j < g_{ij}(b; b)$, box j is inspected if and only if $\underline{v}_{ij} < V_i < \bar{v}_{ij}$.*

The two cutoffs have different economic origins. A sufficiently low realization can terminate search because it is bad news about the common component and therefore about the remaining box. A sufficiently high realization terminates search for the more familiar reason that it creates a strong incumbent. Log-concavity delivers the single-peaked continuation gain; the tail-attribution conditions determine whether these two stopping forces are reached at finite realizations. Gaussian learning satisfies both conditions, as the next section will make explicit.

4.2 Gaussian Components

The previous subsection establishes an interval rule when only one box remains. With several unopened boxes, the continuation decision also depends on how the current observation affects subsequent learning and search. We now specialize to Gaussian components. Posterior variance then depends only on the inspected set, allowing us to verify the posterior-contraction condition in Section 3.2 at every history. The two-sided stopping rule extends to any finite number of heterogeneous boxes.

Assumption 3. *The common and idiosyncratic components satisfy*

$$X \sim \mathcal{N}(m_0, q_0), \quad \varepsilon_i \sim \mathcal{N}(\mu_i, \sigma_i^2), \quad q_0 \geq 0, \quad \sigma_i^2 > 0. \quad (21)$$

The independence assumptions and positive inspection costs remain as specified above. When $q_0 > 0$, these distributions satisfy Assumption 2. We also allow $q_0 = 0$ as the boundary case in which the common component is known and box values are independent.

Proposition 6 (Gaussian updating and optimal search). *Under Assumption 3, consider a state (R, π, b) with $R \neq \emptyset$ and $\pi = \mathcal{N}(m, q)$. Consider a fixed unopened box $i \in R$.*

(i) *The predictive distribution is $Q_i^\pi = \mathcal{N}(m + \mu_i, q + \sigma_i^2)$. After observing $V_i = v$, the posterior becomes*

$$T_i(\pi, v) = \mathcal{N}(m + \kappa_i(q)(v - m - \mu_i), \tau_i(q)), \quad (22)$$

where $\kappa_i(q) = q/(q + \sigma_i^2)$ and $\tau_i(q) = q\sigma_i^2/(q + \sigma_i^2)$.

(ii) *Suppose $q > 0$ and inspecting box i leaves a nonempty set $R' = R \setminus \{i\}$. The post-signal frontier is*

$$r_i(v) = m + \kappa_i(q)(v - m - \mu_i) + \zeta(\tau_i(q), R'). \quad (23)$$

where $\zeta(q, R) = \rho_R(\mathcal{N}(0, q))$ is independent of v . Denote $\Delta_i = r_i(b) - b$. If $\Delta_i \leq 0$, no realization of box i induces further search. If $\Delta_i > 0$, there are unique finite cutoffs

$$\underline{v}_i = b - \frac{\Delta_i}{\kappa_i(q)}, \quad \bar{v}_i = b + \frac{\Delta_i}{1 - \kappa_i(q)}, \quad (24)$$

with $\underline{v}_i < b < \bar{v}_i$, such that the decision maker continues after inspecting i if and only if $\underline{v}_i < V_i < \bar{v}_i$. Otherwise, she stops.

When $q > 0$, beliefs respond to the difference between the observed value and its prediction, $v - (m + \mu_i)$. A positive surprise raises the estimated common component and hence the expected values of all unopened boxes; a negative surprise lowers them. The weight $\kappa_i(q)$ lies strictly

between zero and one, so only part of the surprise is attributed to the common component. This weight is larger when common uncertainty is greater or the inspected box has less idiosyncratic noise. Inspection also reduces posterior variance from q to $\tau_i(q)$, regardless of the realization.

The second part of this Proposition characterizes optimal stopping after inspecting a given box i . It does not by itself identify which box should be inspected next; that choice still requires comparing the continuation payoffs in Section 3.1. Whenever $\Delta_i > 0$, the continuation region is bounded by two finite cutoffs. Sufficiently bad realizations stop search by lowering beliefs about the common component, whereas sufficiently good realizations stop search by creating a strong incumbent. The result follows from the posterior-contraction argument in Section 3.2. Gaussian updating makes the posterior variance $\tau_i(q)$ independent of the realization, and the translation property of the frontier implies that $r_i(v)$ is affine with slope $\kappa_i(q) \in (0, 1)$, verifying condition (PC). Below b , the incumbent remains fixed while the frontier rises with the observation; above b , the incumbent rises one-for-one while the frontier rises more slowly. Thus continuation occurs on a finite interval containing b precisely when $\Delta_i = r_i(b) - b > 0$. As common uncertainty vanishes ($q \downarrow 0$), the stopping frontier satisfies $\zeta(q, R) \rightarrow \max_{i \in R} z_i(0)$, recovering Weitzman's one-sided stopping rule for independent values.

Whenever the continuation interval is nonempty, its asymmetry satisfies

$$\frac{\bar{v}_i - b}{b - \underline{v}_i} = \frac{\kappa_i}{1 - \kappa_i} = \frac{q}{\sigma_i^2}. \quad (25)$$

A larger ratio of common uncertainty to idiosyncratic uncertainty makes the upper side of the interval longer relative to the lower side. The observation then conveys more news about the common component, strengthening both the adverse effect of bad news and the favorable effect of good news on the outlook for unopened boxes. Changes in q or σ_i^2 also change Δ_i , so this ratio alone does not determine how either cutoff moves.

Suppose $q > 0$, the continuation interval in Proposition 6 is nonempty, and the incumbent b comes from a previously inspected box. If $v \leq \underline{v}_i$, the decision maker stops and recalls the incumbent; if $\underline{v}_i < v < b$, she continues searching. Neither realization improves the available payoff, but sufficiently bad news lowers beliefs about the remaining boxes enough to trigger recall. Under independent values, all observations below b leave the incumbent and the distributions of the remaining boxes unchanged, and therefore lead to the same continuation decision.

The gain from inspecting a box combines its direct prize surplus, net of inspection cost, with the expected surplus from subsequent search. The latter includes both further prize opportunities and the value of adapting future inspections to the observation. Fix $q > 0$ and compare boxes with the same mean and cost. A lower σ_i^2 raises $\kappa_i(q)$ and lowers $\tau_i(q)$: the posterior mean responds more strongly to a given surprise, and less uncertainty about X remains after inspection. A higher σ_i^2 , however, increases the predictive variance $q + \sigma_i^2$ and the expected improvement over the incumbent. These informational and prize effects favor different boxes, so signal precision alone does not determine the optimal search order. When $q = 0$, $\kappa_i(0) = \tau_i(0) = 0$, learning disappears, and Weitzman's rule ranks the higher-variance box first.

Identical boxes. Suppose all unopened boxes have common parameters (μ, σ^2, c) . Write $\rho_n(\pi)$ for the reservation frontier with n unopened boxes and $r_n(v) = \rho_{n-1}(T(\pi, v))$ for the frontier

after inspecting one of them, where T is their common posterior updating rule.

Proposition 7. *Fix the current posterior $\pi = \mathcal{N}(m, q)$. Then*

$$\rho_{n+1}(\pi) \geq \rho_n(\pi) \quad (n \geq 1), \quad r_{n+1}(v) \geq r_n(v) \quad (n \geq 2)$$

for every realization v . If $q > 0$ and the continuation interval is nonempty, its cutoffs satisfy

$$\underline{v}_{n+1} \leq \underline{v}_n, \quad \bar{v}_{n+1} \geq \bar{v}_n.$$

The learning premium $\Lambda_n(\pi) = \rho_n(\pi) - \rho_1(\pi)$ is weakly increasing in n , starting from $\Lambda_1(\pi) = 0$. With more boxes remaining, an observation can guide more subsequent inspection decisions, weakly increasing the value of contingent continuation. This raises the reservation frontier and widens the continuation interval. When $q = 0$, the learning premium is zero for every n , so these thresholds are independent of the number of boxes.

Two boxes and Search Order Reversal. We next consider two boxes and ask when greater common uncertainty reverses the independent-value search order. Let $R = \{L, H\}$, with current posterior $\pi = \mathcal{N}(m, q)$ and incumbent b . The boxes have a common idiosyncratic mean μ , a common inspection cost c , and variances $0 < \sigma_L^2 < \sigma_H^2$. We vary q while holding m , b , and the boxes' parameters fixed.

Write $S = \sqrt{\sigma_L^2 + \sigma_H^2}$ and suppose $0 < c < S/\sqrt{2\pi}$. Let ϕ and Φ denote the standard normal density and distribution function, and write $\bar{\Phi} = 1 - \Phi$. Define $a_c > 0$ by

$$c = S[\phi(a_c) - a_c \bar{\Phi}(a_c)], \tag{26}$$

and let

$$\Gamma(c, S) = \frac{S^2}{2c} [(1 + a_c^2) \bar{\Phi}(a_c) - a_c \phi(a_c)]. \tag{27}$$

This threshold determines which inspection is preferred when common uncertainty is sufficiently large.

Proposition 8 (Gaussian search-order reversal). *(i) If $m + \mu - b > \Gamma(c, S)$, search is active for every $q \geq 0$. Box H is uniquely optimal to inspect first at $q = 0$, whereas box L is uniquely optimal for all sufficiently large q . Hence the optimal first inspection reverses at least once as q increases.*

(ii) If $m + \mu - b < \Gamma(c, S)$, box H is uniquely optimal to inspect first for all sufficiently large q .

At $q = 0$, observing one box provides no information about the other, and H has the higher Weitzman reservation value. For $q > 0$, inspecting L leaves less uncertainty about X , since $\tau_L(q) < \tau_H(q)$. Its larger updating weight, $\kappa_L(q) > \kappa_H(q)$, also means that the same negative surprise produces a larger downward revision in the expected value of the remaining box. This information can help the decision maker stop before paying for another inspection. When b lies far below $m + \mu$, observations near or below b are particularly unfavorable surprises. Part (i) identifies how large this gap must be for the informational benefit to make L the preferred first

inspection when q is sufficiently large. If the gap is below $\Gamma(c, S)$, part (ii) instead favors H for sufficiently large q .

The cost restriction concerns the return to a second inspection. After inspecting i , the remaining box j has conditional variance $\sigma_j^2 + \tau_i(q)$, which converges to S^2 . If its conditional mean equals the current incumbent, the expected improvement over the incumbent therefore converges to $S/\sqrt{2\pi}$: a low realization can be rejected, while a high realization improves the payoff. Hence $c < S/\sqrt{2\pi}$ ensures that this inspection is worthwhile for sufficiently large q .

Condition (i) also guarantees that search is worth starting. Since $\Gamma(c, S) > c$, it implies $m + \mu - b > c$, so even a single inspection followed by stopping is profitable. Part (i) guarantees at least one change in the optimal order, but does not establish that the order changes only once. At $m + \mu - b = \Gamma(c, S)$, the result does not determine which box should be inspected first.

4.3 Exponential Components (Preliminary)

Consider a firm testing alternative production methods based on the same technology. Each method delivers the technology's unknown potential payoff less an implementation loss. Testing is costly and reveals a method's payoff, thereby also informing the firm about the prospects of untested methods. The firm can retain an existing method yielding u_0 , which provides a known lower bound on the technology's potential. We assume exponential distributions for the remaining potential above u_0 and for the implementation losses, allowing a tractable characterization of learning and optimal search.

Exponential components provide two further tractable additive environments. The first admits an explicit solution for an arbitrary finite set of heterogeneous boxes: an optimal inspection set can be chosen before any values are observed. The second retains a realization-dependent stopping decision, with two finite cutoffs whenever continuation is possible. These specifications use one-sided component distributions and therefore replace, rather than impose, the full-support assumption used in Section 4.1. Throughout this subsection, $\text{Exp}(a)$ denotes the exponential distribution with rate $a > 0$.

A common upper bound and exponential shortfalls. Suppose

$$X = u_0 + E_0, \quad V_i = X - E_i, \quad E_0 \sim \text{Exp}(a_0), \quad E_i \sim \text{Exp}(\lambda_i), \quad (28)$$

where all E_i , including E_0 , are mutually independent, $a_0, \lambda_i > 0$, and $c_i > 0$. The common component X is an unknown upper bound on every box's value, and E_i is the shortfall of box i from that bound. The lower endpoint of the prior distribution of X equals the outside option u_0 . This alignment is part of the specification.

For a history $h = (S, v_S)$, write

$$a = a_0 + \sum_{j \in S} \lambda_j, \quad b = \max\{u_0, \max_{j \in S} v_j\}, \quad R = I \setminus S.$$

For $T \subseteq R$, let $\Lambda_T = \sum_{i \in T} \lambda_i$ and $C_T = \sum_{i \in T} c_i$, with both sums equal to zero for $T = \emptyset$.

Proposition 9 (Exponential updating and an optimal inspection set). *Under (28), the following statements hold at every feasible history, for any finite number of boxes.*

(i) *The posterior distribution of the common component is*

$$X \mid h \stackrel{d}{=} b + \text{Exp}(a). \quad (29)$$

After inspecting $i \in R$ and observing $V_i = v$, the posterior parameters become $a' = a + \lambda_i$ and $b' = \max\{b, v\}$.

(ii) *The optimal continuation value is*

$$W_R(\pi, b) = b + \max_{T \subseteq R} \left\{ \frac{1}{a} - \frac{1}{a + \Lambda_T} - C_T \right\}, \quad (30)$$

where π is the posterior in (29). Choose a maximizing set T^ with no proper subset that also maximizes the expression. An optimal policy inspects every box in T^* , in any order, and then stops. In particular, for $R \neq \emptyset$, continuation is optimal if and only if*

$$\max_{i \in R} \left\{ \frac{\lambda_i}{a(a + \lambda_i)} - c_i \right\} > 0. \quad (31)$$

(iii) *If the rates λ_i are common, there is an optimal policy that always inspects a cheapest remaining box while (31) holds. If the costs c_i are common, there is an optimal policy that always inspects a remaining box with the highest λ_i while that condition holds.*

The posterior has a particularly useful interpretation: conditional on the history, the distance from the incumbent to the common upper bound is exponential with mean $1/a$. A new record raises the incumbent and the lower endpoint of the posterior by the same amount. An observation below the incumbent leaves both unchanged. In either case, inspection increases the posterior rate by the known amount λ_i . Thus the distribution of $X - b$ after inspection depends on which box was opened, but not on its realized value. Beliefs change, yet those changes create no benefit from making future inspection decisions contingent on the observed values.

The expected improvement from inspecting a single box is

$$\mathbb{E}[(V_i - b)^+ \mid h] = \frac{\lambda_i}{a(a + \lambda_i)} = \frac{1}{a} - \frac{1}{a + \lambda_i}.$$

This gain falls as a increases. Hence, if every available inspection has a nonpositive current net gain, later inspections cannot make a sequence of such inspections profitable. For a fixed inspected set, the expected improvements sum to the difference between the initial and final values of $1/a$, yielding (30). When both costs and rates vary, this formula determines the optimal set jointly. The stopping test in (31) alone does not provide a general priority ranking of boxes.

For identical boxes with rate λ and cost c , the rule is even simpler. After k inspections, open another box, if one remains, exactly when

$$\frac{\lambda}{(a_0 + k\lambda)(a_0 + (k + 1)\lambda)} > c. \quad (32)$$

For example, if $a_0 = \lambda = 1$ and $c = 1/10$, the first three expected improvements are $1/2$, $1/6$, and $1/12$. With at least two available boxes, exactly two inspections are optimal, regardless of the observed values. This provides a contrast with the Gaussian case: correlated values and Bayesian learning can coexist with an optimal inspection plan that is independent of realized values.

A common lower bound and exponential additions. Consider instead the separate specification

$$X = M_0 - E_0, \quad V_i = X + E_i, \quad E_0 \sim \text{Exp}(a_0), \quad E_i \sim \text{Exp}(\lambda_i), \quad (33)$$

again with mutual independence, positive rates, and positive inspection costs. Here $M_0 \in \mathbb{R}$ is a known upper endpoint for the common component, and the outside option $u_0 \in \mathbb{R}$ is unrestricted. The component X is an unknown lower bound on the box values. Following history $h = (S, v_S)$, define

$$a = a_0 + \sum_{j \in S} \lambda_j, \quad M = \min\{M_0, \min_{j \in S} v_j\}.$$

The posterior is $X \mid h \stackrel{d}{=} M - \text{Exp}(a)$. Inspection of box i changes these parameters to $a' = a + \lambda_i$ and $M' = \min\{M, v\}$.

For any nonempty remaining set A , define

$$\zeta_A(a) = \rho_A(\mathcal{L}(-E)), \quad E \sim \text{Exp}(a),$$

where $\mathcal{L}$ denotes a probability law. Translation invariance then gives $\rho_A(\mathcal{L}(M - E)) = M + \zeta_A(a)$. The function ζ_A also depends on the costs and rates of the boxes in A .

Proposition 10 (Two cutoffs with exponential additions). *Under (33), consider a state with posterior $\mathcal{L}(M - \text{Exp}(a))$ and incumbent b . Suppose inspecting i leaves $R' = R \setminus \{i\} \neq \emptyset$, and write $z = \zeta_{R'}(a + \lambda_i)$. The post-signal frontier is*

$$r_i(v) = \min\{M, v\} + z. \quad (34)$$

If $z \leq (b - M)^+$, no realization induces further search. If $z > (b - M)^+$, there are two finite cutoffs,

$$\underline{v}_i = b - z, \quad \bar{v}_i = M + z, \quad (35)$$

with $\underline{v}_i < b < \bar{v}_i$. The decision maker continues after inspecting i if and only if $\underline{v}_i < V_i < \bar{v}_i$.

This model makes the informational role of low observations explicit. Only a realization below M lowers the posterior endpoint. Sufficiently low observations therefore reduce the continuation frontier below the unchanged incumbent and induce stopping. All observations above M generate the same posterior, whereas the incumbent continues to rise when the observed value exceeds b . Sufficiently high observations thus also induce stopping. Between these two regions, further search is optimal whenever the continuation interval is nonempty.

The connection to Section 3.2 is direct: the rule follows by comparing $r_i(v)$ with $\max\{b, v\}$. The frontier in (34) has slope one below M and zero above M . Consequently, it does not

satisfy the uniform strict contraction condition stated there, even though it has two finite cutoffs whenever continuation is possible. The two-cutoff property therefore extends beyond that sufficient condition.

For arbitrary remaining sets, $\zeta_{R'}$ is still a continuation frontier and need not have an elementary expression. The proposition characterizes stopping after a specified inspection; it does not provide a general index for heterogeneous boxes. If all λ_i are equal, however, conditional exchangeability implies that there is an optimal policy that selects a cheapest remaining box whenever search continues. With common costs as well, every remaining box is equivalent.

With just one remaining box j , the cutoff constant is elementary whenever it is positive. Specifically, at posterior rate $A > 0$, if $c_j < A/[\lambda_j(A + \lambda_j)]$, then

$$\zeta_{\{j\}}(A) = \frac{1}{\lambda_j} \log \left(\frac{A}{c_j \lambda_j (A + \lambda_j)} \right) > 0.$$

For example, start with $a_0 = 1$, $M_0 = u_0 = 0$, and two boxes with $\lambda_i = 1$ and $c_i = 1/10$. After the first inspection, the rate is $A = 2$ and the decision maker opens the second box exactly when

$$-\log(20/3) < V_i < \log(20/3).$$

The cutoffs are approximately -1.8971 and 1.8971 .

5 Conclusion

This paper studies how information spillovers shape sequential search. A posterior reservation frontier summarizes optimal stopping for a finite set of interdependent alternatives, while its learning premium measures the additional willingness to search relative to evaluating boxes in isolation. Inspection choice reflects two distinct benefits: obtaining a valuable payoff and learning about the alternatives that remain available. Their relative importance can change with beliefs and the incumbent, so a fixed ranking of alternatives need not be optimal.

The additive models make these forces concrete. Poor observations can justify stopping by lowering expected gains from further search, while favorable observations can justify stopping by improving the incumbent. Log-concavity yields an interval of continuation when one box remains; additional tail restrictions are needed to ensure two finite cutoffs. With positive common uncertainty, Gaussian updating delivers two finite cutoffs whenever continuation is possible, at any finite horizon. In the two-box Gaussian comparison, increasing common uncertainty can also shift the optimal first inspection from a more dispersed alternative to a more informative one under the conditions we identify.

The exponential specifications clarify the role of adaptation. When payoffs are exponential shortfalls from a common upper bound whose exponential prior has its lower endpoint at the outside option, the posterior lower endpoint moves with the incumbent. The distribution of the remaining distance to the common bound then depends on the inspected set but not on realized values, and an optimal inspection set can be chosen in advance. With exponential additions above a common lower bound, a bounded continuation interval arises whenever continuation is possible,

even though the frontier responds one-for-one to observations over part of the observation range. These results distinguish Bayesian updating from the value of adapting inspection decisions and show that strict posterior contraction is sufficient but not necessary for interval rules.

An extension with endogenous prices or information disclosure could use this framework to study how firms influence both the immediate attractiveness of their products and the information those products reveal about other alternatives.

A Proofs for the General Results

All beliefs in a continuation problem with unopened set R are understood to be R -admissible. Conditional statements use the regular posterior versions specified in the model and hold outside null histories. We use the convention $\sup \emptyset = -\infty$ and stop whenever continuation and stopping have equal value.

A.1 Admissibility and the Bellman equation

Let $Y_R = \max_{j \in R} V_j^+$, with $Y_\emptyset = 0$. If π is R -admissible and $i \in R$, the tower property gives

$$\mathbb{E}_\pi[\mathbb{E}_\pi[Y_{R \setminus \{i\}} \mid V_i]] = \mathbb{E}_\pi[Y_{R \setminus \{i\}}] < \infty.$$

Conditional independence identifies the inner expectation with the predictive expectation under $T_i(\pi, V_i)$. Thus this posterior is $R \setminus \{i\}$ -admissible for Q_i^π -almost every observation. Starting from the admissible prior and iterating proves admissibility of the unopened set after almost every history, including histories generated by an adaptive inspection policy.

For any feasible policy α , write J_α for its realized payoff from the current state, excluding sunk costs. Pathwise,

$$b - \sum_{j \in R} c_j \leq J_\alpha \leq b + \left(\max_{j \in R} V_j - b \right)^+. \quad (\text{A.1})$$

The upper bound is integrable under R -admissibility. Consequently every policy payoff is integrable and all continuation values are finite.

Given a history h , the conditional law of the unopened vector is

$$\int_{\Theta} \bigotimes_{j \in R(h)} P_j(dv_j \mid \theta) d\pi_h(\theta).$$

It depends on past observations through π_h , while terminal choice depends on past prizes through $b(h)$. Hence (R, π, b) is sufficient. There are finitely many feasible actions at each stage. Backward induction, with a fixed measurable rule for inspection ties, yields an optimal policy and the

Bellman equation

$$\begin{aligned}
W_\emptyset(\pi, b) &= b, \\
W_R(\pi, b) &= \max\{b, \max_{i \in R} H_{i,R}(\pi, b)\}, \\
H_{i,R}(\pi, b) &= -c_i + \int_{\mathbb{R}} W_{R \setminus \{i\}}(T_i(\pi, v), b \vee v) dQ_i^\pi(v).
\end{aligned} \tag{A.2}$$

A.2 Proof of Proposition 1

Proof. We induct on $|R|$. For $R = \emptyset$, $W_\emptyset(\pi, b) = b$. Suppose the claims hold for every smaller unopened set. Fix $i \in R$ and $b' \geq b$. Since $0 \leq (b' \vee v) - (b \vee v) \leq b' - b$, the induction hypothesis implies

$$0 \leq W_{R \setminus \{i\}}(T_i(\pi, v), b' \vee v) - W_{R \setminus \{i\}}(T_i(\pi, v), b \vee v) \leq b' - b$$

for almost every v . Integration proves that $H_{i,R}(\pi, \cdot)$ is nondecreasing and one-Lipschitz. Taking the finite maximum with the stopping payoff preserves both properties, so they also hold for W_R .

The map $b \mapsto b \vee v$ is convex. Its composition with the convex, nondecreasing continuation value in the induction hypothesis is convex. Integration therefore preserves convexity of $H_{i,R}$, and finite maximization preserves convexity of W_R . Lipschitz continuity implies continuity. Finiteness follows from (A.1). $\square$

A.3 Proof of Theorem 1

Proof. Proposition 1 implies that $H_R(\pi, \cdot) = \max_{i \in R} H_{i,R}(\pi, \cdot)$ is continuous and one-Lipschitz. Thus $G_R(\pi, b) = H_R(\pi, b) - b$ is continuous and nonincreasing in b . Opening any given box i and stopping immediately is feasible, so

$$G_R(\pi, b) \geq \mathbb{E}_\pi[(V_i - b)^+] - c_i. \tag{A.3}$$

As $b \downarrow -\infty$, the expectation tends to infinity by monotone convergence; no negative-moment assumption is needed. Conversely, any policy that continues pays at least $c_{\min} = \min_{i \in R} c_i > 0$ and can select no prize above $b \vee \max_{j \in R} V_j$. Therefore

$$G_R(\pi, b) \leq \mathbb{E}_\pi \left[\left(\max_{j \in R} V_j - b \right)^+ \right] - c_{\min}. \tag{A.4}$$

As $b \uparrow \infty$, the expectation tends to zero by dominated convergence and admissibility. Hence G_R is positive for sufficiently low incumbents and negative for sufficiently high incumbents.

The nonpositive set of G_R is consequently a nonempty closed upper interval with a finite lower endpoint $\rho_R(\pi)$. Below this endpoint, $H_R(\pi, b) > b$; at and above it, $H_R(\pi, b) \leq b$. The stopping convention and (A.2) give both the stated stopping rule and the choice of an inspection maximizing $H_{i,R}(\pi, b)$. $\square$

A.4 Proof of Proposition 2

Proof. If $R \subseteq R'$, every policy available with R remains available with R' by ignoring the added boxes. Thus $W_{R'}(\pi, b) \geq W_R(\pi, b)$. Whenever search is strictly profitable with R , it is strictly profitable with R' , which gives $\rho_{R'}(\pi) \geq \rho_R(\pi)$. For an empty set the same statement holds with the convention $\rho_\emptyset = -\infty$.

Lowering any inspection cost weakly raises the payoff of every fixed policy and leaves the set of feasible policies unchanged. The same inclusion of strict continuation regions yields cost monotonicity. $\square$

A.5 Proof of Theorem 2

Proof. After observing v , continuation is strictly optimal exactly when $b \vee v < r_i(v)$. On $(-\infty, b]$, this condition is $b < r_i(v)$. Condition (PC) makes r_i nondecreasing. On $[b, \infty)$, put $h(v) = r_i(v) - v$. For $v' > v \geq b$,

$$h(v') - h(v) \leq -(1 - \kappa_i)(v' - v) < 0, \quad h(v) \leq h(b) - (1 - \kappa_i)(v - b). \quad (\text{A.5})$$

Thus h is strictly decreasing and tends to $-\infty$.

If $r_i(b) \leq b$, neither branch permits continuation. If $r_i(b) > b$, continuity gives continuation in a neighborhood of b . Equation (A.5) then gives a unique finite upper cutoff $\bar{v}_i > b$ satisfying $r_i(\bar{v}_i) = \bar{v}_i$. Define

$$\underline{v}_i = \sup\{v \leq b : r_i(v) \leq b\}.$$

If this set is empty, $\underline{v}_i = -\infty$ and every $v \leq b$ permits continuation. Otherwise, monotonicity and continuity imply $\underline{v}_i < b$, $r_i(\underline{v}_i) = b$, and $r_i(v) > b$ precisely for $v \in (\underline{v}_i, b]$. This remains true if the equation $r_i(v) = b$ has several solutions: the supremum selects the right endpoint of that level set. Combining the two branches gives continuation exactly when $\underline{v}_i < v < \bar{v}_i$. $\square$

A.6 Proof of Proposition 3

Proof. For an integrable positive part of V_i , the function $C_i(b) = \mathbb{E}_\pi[(V_i - b)^+]$ is finite, continuous, and nonincreasing. It tends to infinity as $b \downarrow -\infty$ and to zero as $b \uparrow \infty$. It is strictly decreasing below the essential supremum of V_i . Since $c_i > 0$, a unique finite $z_i(\pi)$ solves $C_i(z_i(\pi)) = c_i$. By (A.3), every $b < z_i(\pi)$ lies in the strict continuation region. Hence $\rho_R(\pi) \geq \max_{i \in R} z_i(\pi)$. With one unopened box, opening and stopping is the only continuation policy, so equality holds.

For completeness, equality under independence follows directly by induction. Suppose $b \geq \max_{i \in R} z_i(\pi)$. An observation of box i leaves the joint predictive law of the remaining boxes unchanged, and the new incumbent is at least b . By the induction hypothesis, stopping is optimal thereafter. Consequently $H_{i,R}(\pi, b) - b = C_i(b) - c_i \leq 0$ for every i , and stopping is optimal initially. This proves the independent-value equality.

Let i^* and z^* be as in the proposition. Adding and subtracting the post-inspection incumbent

gives

$$H_{i^*,R}(\pi, z^*) - z^* = \mathbb{E}_\pi[W_{R \setminus \{i^*\}}(T_{i^*}(\pi, V_{i^*}), z^* \vee V_{i^*}) - (z^* \vee V_{i^*})], \quad (\text{A.6})$$

because $C_{i^*}(z^*) = c_{i^*}$. The integrand is nonnegative and integrable. Under the stated positive-probability condition, its expectation is strictly positive. Theorem 1 then implies $\rho_R(\pi) > z^*$ and $\Lambda_R(\pi) > 0$. $\square$

A.7 Proof of Proposition 4

Proof. Fix R and b . A feasible policy is a measurable rule assigning an action to each observable history. The class of such rules is the same under every belief. For a fixed rule α , let $J_\alpha(\theta)$ be its expected payoff conditional on the latent state. Its ex ante value is linear in the belief:

$$J_\alpha(\pi) = \int_{\Theta} J_\alpha(\theta) d\pi(\theta), \quad W_R(\pi, b) = \sup_{\alpha} J_\alpha(\pi).$$

The supremum of linear functions is convex, proving the first claim on the convex domain of R -admissible beliefs.

If $\mathbb{E}[\Pi] = \pi$, then Π is R -admissible almost surely, since $\mathbb{E}[\mathbb{E}_\Pi Y_R] = \mathbb{E}_\pi Y_R < \infty$. Moreover, for every fixed policy,

$$\mathbb{E}[W_R(\Pi, b)] \geq \mathbb{E}[J_\alpha(\Pi)] = J_\alpha(\pi).$$

Taking the supremum over α proves the claimed information inequality directly. All expectations are finite by (A.1). Under a Blackwell improvement, the decision maker can reproduce the less informative signal by its garbling and follow the corresponding search policy. Alternatively, apply the same inequality conditional on the less informative signal. Equal experiment costs therefore give the stated comparison of net values. $\square$

A.8 Calculations for Example 1 and the index implication

In Example 1, $b = c = 1/4$, $V_A = \theta$, and $V_B = 2\theta$ with conditional probability $1/2$ and zero otherwise. For $p > 1/2$, the one-box reservation values are

$$z_A = 1 - \frac{1}{4p}, \quad z_B = 2 - \frac{1}{2p} = 2z_A > z_A.$$

After $A = 0$, the remaining prize is zero. After $A = 1$, the expected improvement from B is $1/2 > c$, so B is inspected. If $B = 2$, search stops. If $B = 0$, the posterior is $p/(2-p)$, and inspecting A has gross gain $3p/[4(2-p)] > 1/4$ precisely when $p > 1/2$.

Both orders therefore give expected terminal prize $(1-p)/4 + 3p/2$. Their expected inspection costs are, respectively, $(1+p)/4$ and $(2-p/2)/4$. Hence

$$H_{A,R}(\pi, b) = p, \quad H_{B,R}(\pi, b) = \frac{11p-2}{8}, \quad H_{A,R}(\pi, b) - H_{B,R}(\pi, b) = \frac{2-3p}{8}. \quad (\text{A.7})$$

Since $H_{A,R} > b$ throughout the specified range, the sign determines the optimal first inspection as claimed.

The two relevant posteriors can also occur along an optimal search path within a single fixed problem. Start from $p_0 = 7/10$ and add a preliminary box D , independent of the other boxes conditional on θ , whose values are 0 and $1/8$. Let

$$\mathbb{P}(V_D = 1/8 \mid \theta = 1) = \frac{4}{7}, \quad \mathbb{P}(V_D = 1/8 \mid \theta = 0) = \frac{1}{3}.$$

The two values occur with probability $1/2$ each and induce posteriors $3/5$ and $4/5$, respectively, without changing the incumbent. After opening A , the state is known and D has no value. After $B = 2$, search stops; after $B = 0$, inspecting A is optimal under either possible signal from D , so D again has no value. Thus the best value from starting directly with A or B is $57/80$. Starting with D gives $29/40 - c_D$, which is strictly larger whenever $0 < c_D < 1/80$. Both posteriors therefore arise after an optimal first inspection. At these histories, A and B have the same primitives, remaining set, and incumbent, but opposite strict rankings by (A.7). Fixed box indices with fixed tie-breaking cannot be optimal at both histories.

B Proofs for the Last Unopened Box

B.1 Proof of Proposition 5

Proof. At the current history, let f denote the posterior density of X . It is proportional to $f_X(x) \prod_{k \in S} f_k(v_k - x)$, so it remains positive, continuous, and log-concave. If h is a positive log-concave density and $v_2 > v_1$, then

$$x \mapsto \frac{h(v_2 - x)}{h(v_1 - x)} \quad \text{is nondecreasing.} \quad (\text{B.1})$$

Indeed, for the concave function $\ell = \log h$, the increment $\ell(y + v_2 - v_1) - \ell(y)$ is nonincreasing in y . Substituting $y = v_1 - x$ proves (B.1), without a differentiability assumption.

For $v \leq b$, the posterior density of X and the continuation gain are

$$p_v(x) = \frac{f(x)f_i(v-x)}{\int f(y)f_i(v-y)dy}, \quad g_{ij}(v; b) = \int h_b(x)p_v(x)dx, \quad (\text{B.2})$$

where $h_b(x) = \mathbb{E}[(x + \varepsilon_j - b)^+]$ is strictly increasing, because ε_j has full support. By (B.1), p_v shifts upward in likelihood-ratio order as v increases. This implies that the expectation of h_b is nondecreasing. One direct verification is to write the density ratio as a nondecreasing function L and use $\mathbb{E}_{v_2} h_b - \mathbb{E}_{v_1} h_b = \text{Cov}_{v_1}(h_b, L) \geq 0$.

For $v \geq b$, additivity gives $V_j - v = \varepsilon_j - \varepsilon_i$. The corresponding conditional density and payoff are

$$\tilde{p}_v(e) = \frac{f_i(e)f(v-e)}{\int f_i(y)f(v-y)dy}, \quad g_{ij}(v; b) = \int k(e)\tilde{p}_v(e)de, \quad (\text{B.3})$$

where $k(e) = \mathbb{E}[(\varepsilon_j - e)^+]$ is strictly decreasing. Applying (B.1) to f makes $\tilde{p}_v$ shift upward in

likelihood-ratio order, so this expectation is nonincreasing. On either branch, a nonconstant monotone likelihood ratio yields a strict expectation comparison, since the relevant payoff function is strictly monotone and the densities have full support.

To establish continuity, note that an integrable log-concave density on $\mathbb{R}$ is bounded and has finite absolute first moment. For v in a compact set, the numerator in (B.2) is bounded by an integrable multiple of $f(x)(1 + |x|)$, since f_i is bounded and h_b grows at most linearly. Dominated convergence applies to its numerator and denominator; the latter is continuous and strictly positive. The same argument for (B.3) uses bounded f and the envelope $f_i(e)(1 + |e|)$. Both formulas agree at $v = b$, which proves continuity of g_{ij} .

The strict superlevel set of a continuous function that is nondecreasing up to b and nonincreasing thereafter is an open interval, possibly empty or unbounded. Its maximum is attained at b , so it is empty exactly when $c_j \geq g_{ij}(b; b)$ and otherwise contains b .

For the tail statements following the proposition, suppose $c_j < g_{ij}(b; b)$. If $c_j > g_{ij}^-(b)$, the left tail eventually falls below cost, giving a finite lower endpoint. If $c_j < g_{ij}^-(b)$, the whole left branch is above cost. At equality, the endpoint is finite precisely when the tail level is attained at a finite point; otherwise it is $-\infty$. The upper-tail statements are identical with the inequalities reversed in the realization. In particular, strict monotonicity prevents attainment of a tail limit at a finite point. $\square$

B.2 Calculations for Example 2

Write L for a centered Laplace variable with rate 2, whose density is $e^{-2|x|}$. In the example, the density of X conditional on $X + \varepsilon_i = v$ is proportional to $h_{3,1}(x)h_{1,3}(v - x)$. As $v \rightarrow -\infty$, its limiting density is proportional to $h_{3,1}(x)e^{-x}$, hence is the density of L . Similarly, the conditional density of ε_i as $v \rightarrow \infty$ tends to the density proportional to $h_{1,3}(e)e^e$, which is again that of L . Integrating the exponential pieces on the intervals separated by 0 and v also gives convergence of first absolute moments, so the limits may be taken in the continuation gains.

Since ε_j is an independent copy of L , both $L + \varepsilon_j$ and $\varepsilon_j - L$ have density $\frac{1}{2}(1 + 2|y|)e^{-2|y|}$. Direct integration gives

$$g_{ij}^-(b) = \begin{cases} (b/4 + 3/8)e^{-2b}, & b \geq 0, \\ -b + (-b/4 + 3/8)e^{2b}, & b < 0, \end{cases} \quad g_{ij}^+(b) = \frac{3}{8}. \quad (\text{B.4})$$

For $b \geq 0$, evaluating (B.3) at $v = b$ further gives

$$g_{ij}(b; b) = \frac{3/8 - e^{-4b}/32}{1 - e^{-2b}/3}. \quad (\text{B.5})$$

For example, the unnormalized denominator is $e^{-b} - e^{-3b}/3$, and the numerator is $3e^{-b}/8 - e^{-5b}/32$; the common density constants cancel. Thus at $b = 1/4$ the two tail gains are $(7/16)e^{-1/2} \simeq 0.265357$ and 0.375 , while the peak is approximately 0.455619 . At $b = -1/4$, the lower tail gain becomes $1/4 + (7/16)e^{-1/2} \simeq 0.515357$, while the upper tail remains 0.375 . Proposition 5 then yields all five continuation-set shapes described in the example.

B.3 Proof of Corollary 1

Proof. As $v \rightarrow -\infty$, the assumed conditional convergence of X to $-\infty$, together with the fixed independent distribution of ε_j , implies $(X + \varepsilon_j - b)^+ \rightarrow 0$ in conditional probability. Uniform integrability of these positive parts gives convergence of expectations, so $g_{ij}^-(b) = 0$. As $v \rightarrow \infty$, conditional convergence of ε_i to $+\infty$ similarly implies $(\varepsilon_j - \varepsilon_i)^+ \rightarrow 0$ in conditional probability. The corresponding uniform-integrability assumption gives $g_{ij}^+(b) = 0$. If $0 < c_j < g_{ij}(b; b)$, continuity and Proposition 5 therefore yield a nonempty continuation interval with both endpoints finite. Its endpoints are excluded under the stopping convention. $\square$

C Proofs for Gaussian Components

C.1 Proof of Proposition 6

Proof. Updating. The pair (X, V_i) is jointly normal, with means $(m, m + \mu_i)$, variances $(q, q + \sigma_i^2)$, and covariance q . The conditional-normal formula gives

$$X \mid V_i = v \sim \mathcal{N}\left(m + \frac{q}{q + \sigma_i^2}(v - m - \mu_i), q - \frac{q^2}{q + \sigma_i^2}\right),$$

which proves part (i), including the degenerate case $q = 0$.

Translation and stopping. The following translation property holds in any additive model. If $\pi_a = \mathcal{L}(X + a)$ for $X \sim \pi$, shifting all future observations and the incumbent by a gives a bijection between feasible policies, with each terminal payoff increased by a and all costs unchanged. Therefore

$$W_R(\pi_a, b + a) = a + W_R(\pi, b), \quad H_{i,R}(\pi_a, b + a) = a + H_{i,R}(\pi, b), \quad \rho_R(\pi_a) = a + \rho_R(\pi). \quad (\text{C.1})$$

In particular, $\rho_R(\mathcal{N}(m, q)) = m + \zeta(q, R)$. Applying this identity after the normal update yields

$$r_i(v) = m + \kappa_i(q)(v - m - \mu_i) + \zeta(\tau_i(q), R').$$

For $q > 0$, its slope $\kappa_i(q)$ is strictly between zero and one. Writing $\Delta_i = r_i(b) - b$, the difference between the post-signal frontier and the new incumbent is

$$r_i(v) - (b \vee v) = \begin{cases} \Delta_i + \kappa_i(q)(v - b), & v \leq b, \\ \Delta_i - (1 - \kappa_i(q))(v - b), & v \geq b. \end{cases} \quad (\text{C.2})$$

Its maximum is Δ_i . If this is nonpositive, there is no continuation. Otherwise both branches have unique finite zeros, $\underline{v}_i = b - \Delta_i/\kappa_i(q)$ and $\bar{v}_i = b + \Delta_i/(1 - \kappa_i(q))$, and continuation is strictly optimal exactly between them. This proves part (ii). $\square$

The cutoff-distance ratio stated after the proposition follows immediately:

$$\frac{\bar{v}_i - b}{b - \underline{v}_i} = \frac{\kappa_i(q)}{1 - \kappa_i(q)} = \frac{q}{\sigma_i^2}.$$

If a previous box provides the incumbent, realizations at or below $\underline{v}_i$ induce stopping and recall, whereas those in $(\underline{v}_i, b)$ induce continuation. Under independence, every observation below b leaves both the incumbent and the remaining predictive law unchanged, so this distinction cannot occur.

C.2 The reduced recursion and the independent-value limit

This subsection establishes the limit stated after Proposition 6. Put $w_R(q, d) = W_R(\mathcal{N}(0, q), d)$, so $W_R(\mathcal{N}(m, q), b) = m + w_R(q, b - m)$ by (C.1). With $Y \sim \mathcal{N}(0, 1)$, define

$$Z_i(q) = \sqrt{q + \sigma_i^2} Y, \quad D_i(q, d, Y) = \max\{d, \mu_i + Z_i(q)\} - \kappa_i(q) Z_i(q).$$

The new mean is $m + \kappa_i(q) Z_i(q)$ and the new incumbent-minus-mean gap is $D_i(q, d, Y)$. Its mean shift has expectation zero. Consequently

$$\begin{aligned} w_\emptyset(q, d) &= d, \\ A_{i,R}(q, d) &= -c_i + \mathbb{E}[w_{R \setminus \{i\}}(\tau_i(q), D_i(q, d, Y))], \\ w_R(q, d) &= \max\{d, \max_{i \in R} A_{i,R}(q, d)\}. \end{aligned} \tag{C.3}$$

This supplies an exact recursion for the frontier used in Proposition 6.

We claim that $w_R(q, \cdot)$ and $A_{i,R}(q, \cdot)$ converge locally uniformly to their $q = 0$ counterparts as $q \downarrow 0$. Induct on $|R|$. The assertion is immediate for the empty set. On a bounded interval of q , the Gaussian payoff bound gives

$$|w_R(q, d)| \leq |d| + C_R$$

for a constant independent of q and d . For d in a compact set, $D_i(q, d, Y)$ is bounded in absolute value by $C(1 + |Y|)$. Moreover, $\tau_i(q) \rightarrow 0$, $\kappa_i(q) \rightarrow 0$, and

$$D_i(q, d, Y) \longrightarrow \max\{d, \mu_i + \sigma_i Y\},$$

uniformly over that compact set for each fixed Y . The induction hypothesis, the one-Lipschitz property in the gap, and the integrable linear envelope justify dominated convergence in (C.3), uniformly in d on compact sets. Finite maximization preserves this convergence.

Write $z_i(0)$ for the one-box reservation value of $\mathcal{N}(\mu_i, \sigma_i^2)$ and $z^0 = \max_{i \in R} z_i(0)$. Define $G_R(q, d) = \max_{i \in R} A_{i,R}(q, d) - d$. The preceding argument gives local uniform convergence of G_R . By Proposition 3, $\zeta(0, R) = z^0$. For $d < z^0$, opening a box with reservation value above d and then stopping is strictly profitable, so $G_R(0, d) > 0$. For $d > z^0$, the incumbent after every first inspection already exceeds all remaining reservation values. Thus

$$A_{i,R}(0, d) - d = \mathbb{E}[(\mu_i + \sigma_i Y - d)^+] - c_i < 0$$

for every i , giving $G_R(0, d) < 0$. For any $\epsilon > 0$, local uniform convergence preserves these signs at $z^0 - \epsilon$ and $z^0 + \epsilon$ for sufficiently small q . Theorem 1 then sandwiches the frontier between those

points, proving

$$\lim_{q \downarrow 0} \zeta(q, R) = \max_{i \in R} z_i(0). \quad (\text{C.4})$$

At $q = 0$, further search after inspecting i occurs exactly when $b \vee v < m + \max_{j \in R'} z_j(0)$. If $b < m + \max_{j \in R'} z_j(0)$, the Gaussian cutoff formulas and (C.4) also give $\underline{v}_i \rightarrow -\infty$ and $\bar{v}_i \rightarrow m + \max_{j \in R'} z_j(0)$.

C.3 Proof of Proposition 7

Proof. An additional identical unopened box expands the feasible set of policies, so Proposition 2 gives $\rho_{n+1}(\pi) \geq \rho_n(\pi)$. Applying the same comparison at the common posterior $T(\pi, v)$ gives $r_{n+1}(v) = \rho_n(T(\pi, v)) \geq \rho_{n-1}(T(\pi, v)) = r_n(v)$.

For $q > 0$, every r_n has the same slope $\kappa(q)$, while its intercept is weakly increasing in n . Therefore $\Delta_{n+1} = r_{n+1}(b) - b \geq \Delta_n = r_n(b) - b$. If the interval for n boxes is nonempty, so is that for $n + 1$ boxes, and Proposition 6 gives

$$\underline{v}_{n+1} = b - \frac{\Delta_{n+1}}{\kappa(q)} \leq \underline{v}_n, \quad \bar{v}_{n+1} = b + \frac{\Delta_{n+1}}{1 - \kappa(q)} \geq \bar{v}_n.$$

The common one-box reservation value equals $\rho_1(\pi)$ and does not change with n . Thus $\Lambda_n(\pi) = \rho_n(\pi) - \rho_1(\pi)$ is weakly increasing and starts at zero. At $q = 0$, Proposition 3 gives $\rho_n(\pi) = \rho_1(\pi)$ for every n , and the post-signal frontiers are likewise independent of the number of remaining boxes. $\square$

C.4 Proof of Proposition 8

Proof. Set $\bar{m} = m + \mu$, $d = b - \bar{m}$, and $\nu_i = \sigma_i^2$. Thus $S^2 = \nu_L + \nu_H$ and the gap in the proposition is $-d$. Define

$$h(a) = \phi(a) - a\bar{\Phi}(a), \quad \ell(a) = \frac{\phi(a)}{\bar{\Phi}(a)}, \quad \bar{\Phi}(a) = 1 - \Phi(a).$$

The function h is strictly decreasing, with derivative $h'(a) = -\bar{\Phi}(a)$.

Step 1: an exact comparison of the two orders. Let j denote the other box, let $B_i = b \vee V_i$, and put $g_i(V_i; q) = \mathbb{E}[(V_j - B_i)^+ | V_i]$. Temporarily let the second inspection cost t , holding the first inspection cost at c . The value conditional on starting with i is

$$U_i(q; t) = -c + \mathbb{E}[B_i + (g_i(V_i; q) - t)^+].$$

At $t = 0$, both orders yield $-c + \mathbb{E}[\max\{b, V_L, V_H\}]$. If $P_i(q, t) = \mathbb{P}(g_i(V_i; q) > t)$, Tonelli's theorem and $x - (x - c)^+ = \int_0^c \mathbf{1}\{x > t\} dt$ for $x \geq 0$ give

$$U_H(q; c) - U_L(q; c) = \int_0^c [P_L(q, t) - P_H(q, t)] dt. \quad (\text{C.5})$$

For $q > 0$, the predictive standard deviation of the second box after inspecting i is

$$s_i(q) = \left(\nu_j + \frac{q\nu_i}{q + \nu_i} \right)^{1/2} = \left(S^2 - \frac{\nu_i^2}{q + \nu_i} \right)^{1/2}.$$

For $t > 0$, define $a_i(q, t)$ and $z_i(q, t)$ by $t = s_i(q)h(a_i(q, t))$ and $z_i(q, t) = s_i(q)a_i(q, t)$. Here $z_i(q, t)$ is the reservation premium of the *remaining* box after starting with i . The normal call formula and Proposition 6 give

$$\begin{aligned} P_i(q, t) &= [\Phi(\alpha_i^+(q, t)) - \Phi(\alpha_i^-(q, t))]^+, \\ \alpha_i^-(q, t) &= \frac{(d - z_i(q, t))\sqrt{q + \nu_i}}{q}, \quad \alpha_i^+(q, t) = \frac{z_i(q, t)\sqrt{q + \nu_i}}{\nu_i}. \end{aligned} \quad (\text{C.6})$$

The positive part covers an empty continuation interval; the value at $t = 0$ is immaterial to the integral.

Step 2: the large- q expansion. For $0 < t \leq c$, define $a(t)$ by $t = Sh(a(t))$ and put $z(t) = Sa(t)$. The cost restriction implies $a(t) \geq a(c) = a_c > 0$. If $z(s, t)$ solves $t = sh(z/s)$, implicit differentiation gives

$$\frac{\partial z}{\partial s} = \ell(a), \quad \frac{\partial^2 z}{\partial s^2} = \frac{\ell(a)(\ell(a) - a)^2}{s}, \quad a = z/s. \quad (\text{C.7})$$

Since $s_i(q) = S - \nu_i^2/(2Sq) + O(q^{-2})$, for each fixed $t > 0$,

$$z_i(q, t) = Sa(t) - \frac{\nu_i^2}{2Sq} \ell(a(t)) + O(q^{-2}).$$

Substituting into the lower standardized cutoff yields

$$\alpha_i^-(q, t) = \frac{d - Sa(t)}{\sqrt{q}} + \frac{1}{q^{3/2}} \left[\frac{\nu_i}{2} (d - Sa(t)) + \frac{\nu_i^2}{2S} \ell(a(t)) \right] + O(q^{-5/2}). \quad (\text{C.8})$$

Using $\nu_H + \nu_L = S^2$, subtraction gives

$$q^{3/2}(\alpha_H^- - \alpha_L^-) \longrightarrow \frac{\nu_H - \nu_L}{2} [d + S(\ell(a(t)) - a(t))].$$

Both lower standardized cutoffs tend to zero. The upper standardized cutoffs tend to $+\infty$ at rate $\sqrt{q}$, since $z(t) > 0$. For sufficiently large q , the continuation intervals are nonempty. Consequently the mean value theorem and the Gaussian tail bound imply

$$q^{3/2}[P_L(q, t) - P_H(q, t)] \longrightarrow \frac{\nu_H - \nu_L}{2\sqrt{2\pi}} [d + S(\ell(a(t)) - a(t))]. \quad (\text{C.9})$$

Step 3: justification of the cost integral and its evaluation. The endpoint $t = 0$ requires care because $a(t) \rightarrow \infty$. Set $t_q = Sh(q^{1/4})$. On $[t_q, c]$ we have $a_c \leq a(t) \leq q^{1/4}$. For $a > 0$, the normal

tail inequalities give $0 < \ell(a) - a < 1/a$. They follow, for example, by integration by parts from

$$\frac{a}{1+a^2}\phi(a) < \bar{\Phi}(a) < \frac{\phi(a)}{a}.$$

Together with (C.7), these bounds show that $\partial^2 z / \partial s^2$ is bounded for s near S and the associated a bounded away from zero. Taylor expansion therefore gives, uniformly on $[t_q, c]$,

$$z_i(q, t) = Sa(t) - \frac{\nu_i^2}{2Sq}\ell(a(t)) + O\left(\frac{1+a(t)}{q^2}\right).$$

The remainder in (C.8) is accordingly $O((1+a(t))q^{-5/2})$, uniformly on this interval. Its product with $q^{3/2}$ tends uniformly to zero. Also $\sup_{t \in [t_q, c]} |\alpha_i^-(q, t)| \rightarrow 0$, and the coefficient in (C.9) is bounded because $\ell(a) - a < 1/a_c$. The lower-tail contribution after multiplication by $q^{3/2}$ is thus uniformly bounded.

The premiums $z_i(q, t)$ are bounded below by $z_i(q, c)$, which tends to $Sc_c > 0$. Hence the upper cutoffs satisfy $\alpha_i^+(q, t) \geq K\sqrt{q}$ for a fixed $K > 0$, uniformly on $[t_q, c]$. Their tail contributions vanish uniformly. The same lower bound on premiums implies nonempty continuation intervals, because their condition is $z_i(q, t) > \nu_i d / (q + \nu_i)$. Therefore $q^{3/2}|P_L(q, t) - P_H(q, t)|$ is uniformly bounded on $[t_q, c]$. On $(0, t_q)$ its integral is bounded by $q^{3/2}t_q$, which tends to zero since $h(q^{1/4}) \leq \phi(q^{1/4})$. Dominated convergence now justifies integrating (C.9) in (C.5).

Finally, substituting $t = Sh(a)$ gives

$$\begin{aligned} \int_0^c S[\ell(a(t)) - a(t)] dt &= S^2 \int_{a_c}^\infty [\phi(a) - a\bar{\Phi}(a)] da \\ &= \frac{S^2}{2} [(1+a_c^2)\bar{\Phi}(a_c) - a_c\phi(a_c)] = c\Gamma(c, S). \end{aligned} \tag{C.10}$$

The second equality follows by integration by parts. Thus the exact asymptotic comparison is

$$\lim_{q \rightarrow \infty} q^{3/2}[U_H(q; c) - U_L(q; c)] = \frac{(\nu_H - \nu_L)c}{2\sqrt{2\pi}}[\Gamma(c, S) - (m + \mu - b)]. \tag{C.11}$$

This proves both strict large- q rankings whenever the bracket is nonzero. Search is active for all sufficiently large q , since the one-box gross gain diverges as the predictive standard deviation grows.

Step 4: activity and reversal in part (i). We first verify $\Gamma(c, S) > c$. For $Y \sim \mathcal{N}(0, 1)$, let $Z = (Y - a_c)^+$. Then

$$c = S\mathbb{E}Z, \quad c\Gamma(c, S) = \frac{S^2}{2}\mathbb{E}Z^2.$$

Cauchy-Schwarz gives $(\mathbb{E}Z)^2 \leq \mathbb{P}(Y > a_c)\mathbb{E}Z^2$. Since $a_c > 0$ implies $\mathbb{P}(Y > a_c) < 1/2$, it follows that $\mathbb{E}Z^2 > 2(\mathbb{E}Z)^2$ and hence $\Gamma(c, S) > c$. Under part (i), $m + \mu - b > c$. For every $q \geq 0$ and either box,

$$\mathbb{E}[(V_i - b)^+] \geq \mathbb{E}[V_i - b] = m + \mu - b > c,$$

so search is active throughout.

At $q = 0$, the boxes are independent. For a normal variable with fixed mean, the call value at a finite incumbent has derivative $\phi((b - \bar{m})/s) > 0$ with respect to its standard deviation s . Thus the full one-box reservation values satisfy $r_H^0 > r_L^0$. For clarity, strictness of the first-inspection ranking can be verified without a tie-breaking assumption. Put $Y_i = \min\{V_i, r_i^0\}$. The one-box identity $W_{\{i\}}(x) = \mathbb{E}[\max\{x, Y_i\}]$ follows from $\mathbb{E}[(V_i - r_i^0)^+] = c$. Since $b < r_H^0$, applying this identity to the second inspection gives

$$U_H(0; c) - U_L(0; c) = \mathbb{E}[\min\{(V_L - r_L^0)^+, (\max\{b, Y_H\} - r_L^0)^+\}] > 0.$$

The strict inequality uses independence, full normal support, and $r_H^0 > r_L^0$. Hence H is uniquely optimal at $q = 0$, while (C.11) makes L uniquely optimal for all sufficiently large q under part (i).

The difference $U_H(q; c) - U_L(q; c)$ is continuous in $q \geq 0$. This follows either from the two-box specialization of (C.3) or from the conditional normal call formula: on any compact interval of q , coupling $V_i = \bar{m} + \sqrt{q + \nu_i} Y$ gives continuous integrands with an integrable linear envelope. The opposite strict rankings therefore imply at least one reversal as q increases. The argument neither establishes a unique reversal nor determines the ranking when $m + \mu - b = \Gamma(c, S)$. $\square$

D Proofs for Exponential Components

D.1 Proof of Proposition 9

Proof. Part (i). Under the shortfall specification, the prior density is proportional to $e^{-a_0 x} \mathbf{1}\{x \geq u_0\}$. An observation $V_j = v_j$ has likelihood $\lambda_j e^{-\lambda_j(x - v_j)} \mathbf{1}\{x \geq v_j\}$. Multiplying the prior and all observed likelihoods gives

$$f_{X|h}(x) = a e^{-a(x-b)} \mathbf{1}\{x \geq b\}, \quad a = a_0 + \sum_{j \in S} \lambda_j, \quad b = \max\{u_0, \max_{j \in S} v_j\}. \quad (\text{D.1})$$

Thus $X | h \stackrel{d}{=} b + \text{Exp}(a)$. One further likelihood adds λ_i to the rate and replaces the lower endpoint by $b \vee v$, proving the update.

Part (ii). Given the current history, write $X = b + E$ with $E \sim \text{Exp}(a)$, independent of the unopened shortfalls. For box i , Tonelli's theorem gives the expected improvement

$$\begin{aligned} k_i(a) &= \mathbb{E}[(E - E_i)^+] = \int_0^\infty \mathbb{P}(E - E_i > t) dt \\ &= \int_0^\infty e^{-at} \mathbb{E}[e^{-aE_i}] dt = \frac{\lambda_i}{a(a + \lambda_i)} = \frac{1}{a} - \frac{1}{a + \lambda_i}. \end{aligned} \quad (\text{D.2})$$

Translation invariance makes the continuation surplus independent of b ; denote it by $F(a, R) = W_R(\mathcal{L}(b + \text{Exp}(a)), b) - b$. After inspection, part (i) keeps the same aligned state structure, and the future surplus is $F(a + \lambda_i, R \setminus \{i\})$ for every realization. The Bellman equation therefore becomes

$$F(a, R) = \max \left\{ 0, \max_{i \in R} [k_i(a) - c_i + F(a + \lambda_i, R \setminus \{i\})] \right\}, \quad F(a, \emptyset) = 0. \quad (\text{D.3})$$

Induction solves this finite deterministic recursion. For any first box i and subsequent inspected set $U \subseteq R \setminus \{i\}$, the sum of the gains telescopes to

$$\left(\frac{1}{a} - \frac{1}{a + \lambda_i} - c_i\right) + \left(\frac{1}{a + \lambda_i} - \frac{1}{a + \lambda_i + \Lambda_U} - C_U\right) = \frac{1}{a} - \frac{1}{a + \Lambda_{U \cup \{i\}}} - C_{U \cup \{i\}}.$$

Maximizing over the first box, the subsequent set, and stopping gives

$$F(a, R) = \max_{T \subseteq R} \left\{ \frac{1}{a} - \frac{1}{a + \Lambda_T} - C_T \right\}. \quad (\text{D.4})$$

This argument optimizes over adaptive policies through the original Bellman equation, so it also proves that adaptation to realized values cannot improve upon the best fixed set.

Choose a maximizing set T^* with no proper maximizing subset, and fix any order of its elements. For a proper prefix $P \subsetneq T^*$, the objective for T^* strictly exceeds that for P . Their difference is exactly the surplus from inspecting $T^* \setminus P$ at rate $a + \Lambda_P$, so continuation is strictly profitable at every such prefix. Moreover, a better continuation set at that state could be appended to P to improve the original objective, a contradiction. After T^* is exhausted no further set can have positive surplus, by the same argument. Thus any order within T^* , followed by stopping, is optimal and respects the stopping convention. If the maximum surplus is zero, the minimal maximizing set is empty.

Finally, $k_i(a)$ is decreasing in a . If $k_i(a) - c_i \leq 0$ for every remaining box, each term along any deterministic inspection sequence is nonpositive, since its rate is at least a . Equation (D.4) then gives $F(a, R) = 0$. Conversely, a positive singleton gain makes $F(a, R) > 0$. This proves the stopping test in part (ii).

Part (iii). If all rates equal λ , the gross gain of a k -element set depends only on k , so a best such set consists of the k cheapest boxes. List their costs as $c_{(1)} \leq c_{(2)} \leq \dots$. The incremental objective from adding the k th box is

$$\frac{\lambda}{(a + (k-1)\lambda)(a + k\lambda)} - c_{(k)},$$

which is decreasing in k . Hence a minimal maximizing prefix contains exactly the successive strictly positive increments. This is the stated cheapest-box policy and stopping test.

If all costs equal c , a best k -element set maximizes its total rate and therefore consists of the k largest rates. List them as $\lambda_{(1)} \geq \lambda_{(2)} \geq \dots$, and put $a_{k-1} = a + \sum_{r < k} \lambda_{(r)}$. Its k th increment is $\lambda_{(k)} / [a_{k-1}(a_{k-1} + \lambda_{(k)})] - c$. The gross gain increases in its box rate and decreases in the current posterior rate, so these increments are decreasing in k . The same prefix argument proves the highest-rate policy. $\square$

For identical boxes, the posterior rate after k inspections is $a_0 + k\lambda$. Substitution in (D.2) gives the rule stated after Proposition 9. At $a_0 = \lambda = 1$ and $c = 1/10$, the first two net gains are positive and the third is negative, so exactly two boxes are inspected whenever at least two are available.

D.2 Proof of Proposition 10

Proof. Under the additions specification, the prior density is proportional to $e^{a_0 x} \mathbf{1}\{x \leq M_0\}$. The likelihood of $V_j = v_j$ is $\lambda_j e^{-\lambda_j(v_j - x)} \mathbf{1}\{x \leq v_j\}$. Consequently

$$f_{X|h}(x) = a e^{a(x-M)} \mathbf{1}\{x \leq M\}, \quad a = a_0 + \sum_{j \in S} \lambda_j, \quad M = \min\{M_0, \min_{j \in S} v_j\}. \quad (\text{D.5})$$

Thus $X | h \stackrel{d}{=} M - \text{Exp}(a)$. A further observation changes the rate to $a + \lambda_i$ and the endpoint to $M \wedge v$. Applying (C.1) gives

$$r_i(v) = M \wedge v + \zeta_{R'}(a + \lambda_i) = M \wedge v + z.$$

Theorem 1 therefore prescribes continuation exactly when $b \vee v - M \wedge v < z$. The identity

$$b \vee v - M \wedge v = \max\{b - M, b - v, v - M, 0\} \quad (\text{D.6})$$

shows that this is equivalent to the three strict inequalities

$$z > (b - M)^+, \quad v > b - z, \quad v < M + z.$$

If the first fails, no realization permits continuation. If it holds, $z > 0$ and $M + z > b$, so the finite endpoints satisfy $b - z < b < M + z$. The continuation interval is exactly $(b - z, M + z)$, with stopping at both endpoints. $\square$

D.3 The one-box formula and the equal-rate ordering

With a single remaining box j at posterior $-\text{Exp}(A)$, its value is $E_j - E$ with independent exponential variables of rates λ_j and A . For an incumbent $d \geq 0$,

$$\mathbb{E}[(E_j - E - d)^+] = \frac{e^{-\lambda_j d}}{\lambda_j} \mathbb{E}[e^{-\lambda_j E}] = \frac{A}{\lambda_j(A + \lambda_j)} e^{-\lambda_j d}. \quad (\text{D.7})$$

The one-box reservation value is positive precisely when the call value at zero exceeds c_j . In that case, solving (D.7) for equality with c_j yields

$$\zeta_{\{j\}}(A) = \frac{1}{\lambda_j} \log \left(\frac{A}{c_j \lambda_j (A + \lambda_j)} \right) > 0.$$

For the example in the text, the first inspection leaves $A = 2$ and $z = \log(20/3)$. With $M = b = 0$, Proposition 10 gives $(-\log(20/3), \log(20/3))$, whose endpoints are approximately ± 1.897120 .

To justify the cheapest-box statement when all remaining rates are equal, consider a policy that starts with a more expensive box i while a cheaper box k is available. Interchange the labels i and k throughout this policy, including the labels of their observations. Conditional exchangeability preserves the joint distribution of all observations, decisions, and terminal prizes. If the original policy inspects both boxes, its total cost is unchanged. If it inspects only i , the modified policy instead inspects only k and saves $c_i - c_k$. The modified policy therefore has weakly

greater value and starts with the cheaper box. Applying this argument and backward induction at each continuation state yields an optimal policy that always selects a cheapest remaining box. This argument requires equality of the conditional payoff laws; it does not assert a general priority ranking when rates and costs both differ.

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